

```
In [97]: import seaborn as sns
sns.set(style="whitegrid")
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')
```

```
In [98]: fifa_data = pd.read_csv(r'D:\fsds\materials\class materials\june\13th- Seaborn,
fifa_data.head()
```

```
Out[98]:
```

	Unnamed: 0	ID	Name	Age	Photo	Nation
--	------------	----	------	-----	-------	--------

0	0	158023	L. Messi	31	https://cdn.sofifa.org/players/4/19/158023.png	Arg
---	---	--------	----------	----	---	-----

1	1	20801	Cristiano Ronaldo	33	https://cdn.sofifa.org/players/4/19/20801.png	Po
---	---	-------	-------------------	----	---	----

2	2	190871	Neymar Jr	26	https://cdn.sofifa.org/players/4/19/190871.png	
---	---	--------	-----------	----	---	--

3	3	193080	De Gea	27	https://cdn.sofifa.org/players/4/19/193080.png	
---	---	--------	--------	----	---	--

4	4	192985	K. De Bruyne	27	https://cdn.sofifa.org/players/4/19/192985.png	Be
---	---	--------	--------------	----	---	----

5 rows × 89 columns



```
In [99]: fifa_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 18207 entries, 0 to 18206
```

```
Data columns (total 89 columns):
```

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	18207 non-null	int64
1	ID	18207 non-null	int64
2	Name	18207 non-null	object
3	Age	18207 non-null	int64
4	Photo	18207 non-null	object
5	Nationality	18207 non-null	object
6	Flag	18207 non-null	object
7	Overall	18207 non-null	int64
8	Potential	18207 non-null	int64
9	Club	17966 non-null	object
10	Club Logo	18207 non-null	object
11	Value	18207 non-null	object
12	Wage	18207 non-null	object
13	Special	18207 non-null	int64
14	Preferred Foot	18159 non-null	object
15	International Reputation	18159 non-null	float64
16	Weak Foot	18159 non-null	float64
17	Skill Moves	18159 non-null	float64
18	Work Rate	18159 non-null	object
19	Body Type	18159 non-null	object
20	Real Face	18159 non-null	object
21	Position	18147 non-null	object
22	Jersey Number	18147 non-null	float64
23	Joined	16654 non-null	object
24	Loaned From	1264 non-null	object
25	Contract Valid Until	17918 non-null	object
26	Height	18159 non-null	object
27	Weight	18159 non-null	object
28	LS	16122 non-null	object
29	ST	16122 non-null	object
30	RS	16122 non-null	object
31	LW	16122 non-null	object
32	LF	16122 non-null	object
33	CF	16122 non-null	object
34	RF	16122 non-null	object
35	RW	16122 non-null	object
36	LAM	16122 non-null	object
37	CAM	16122 non-null	object
38	RAM	16122 non-null	object
39	LM	16122 non-null	object
40	LCM	16122 non-null	object
41	CM	16122 non-null	object
42	RCM	16122 non-null	object
43	RM	16122 non-null	object
44	LWB	16122 non-null	object
45	LDM	16122 non-null	object
46	CDM	16122 non-null	object
47	RDM	16122 non-null	object
48	RWB	16122 non-null	object
49	LB	16122 non-null	object
50	LCB	16122 non-null	object
51	CB	16122 non-null	object
52	RCB	16122 non-null	object
53	RB	16122 non-null	object
54	Crossing	18159 non-null	float64

```

55 Finishing 18159 non-null float64
56 HeadingAccuracy 18159 non-null float64
57 ShortPassing 18159 non-null float64
58 Volleys 18159 non-null float64
59 Dribbling 18159 non-null float64
60 Curve 18159 non-null float64
61 FKAccuracy 18159 non-null float64
62 LongPassing 18159 non-null float64
63 BallControl 18159 non-null float64
64 Acceleration 18159 non-null float64
65 SprintSpeed 18159 non-null float64
66 Agility 18159 non-null float64
67 Reactions 18159 non-null float64
68 Balance 18159 non-null float64
69 ShotPower 18159 non-null float64
70 Jumping 18159 non-null float64
71 Stamina 18159 non-null float64
72 Strength 18159 non-null float64
73 LongShots 18159 non-null float64
74 Aggression 18159 non-null float64
75 Interceptions 18159 non-null float64
76 Positioning 18159 non-null float64
77 Vision 18159 non-null float64
78 Penalties 18159 non-null float64
79 Composure 18159 non-null float64
80 Marking 18159 non-null float64
81 StandingTackle 18159 non-null float64
82 SlidingTackle 18159 non-null float64
83 GKDividing 18159 non-null float64
84 GKHandling 18159 non-null float64
85 GK Kicking 18159 non-null float64
86 GKPositioning 18159 non-null float64
87 GKReflexes 18159 non-null float64
88 Release Clause 16643 non-null object
dtypes: float64(38), int64(6), object(45)
memory usage: 12.4+ MB

```

```
In [100... fifa_data['Nationality'].value_counts()
```

```

Out[100... Nationality
England 1662
Germany 1198
Spain 1072
Argentina 937
France 914
...
New Caledonia 1
Fiji 1
São Tomé & Príncipe 1
United Arab Emirates 1
Botswana 1
Name: count, Length: 164, dtype: int64

```

```
In [101... fifa_data['Club'].value_counts()
```

```
Out[101...] Club
FC Barcelona      33
Valencia CF       33
Fortuna Düsseldorf 33
Cardiff City      33
Rayo Vallecano    33
..
América FC (Minas Gerais) 20
Fluminense        20
Sligo Rovers      19
Limerick FC       19
Derry City        18
Name: count, Length: 651, dtype: int64
```

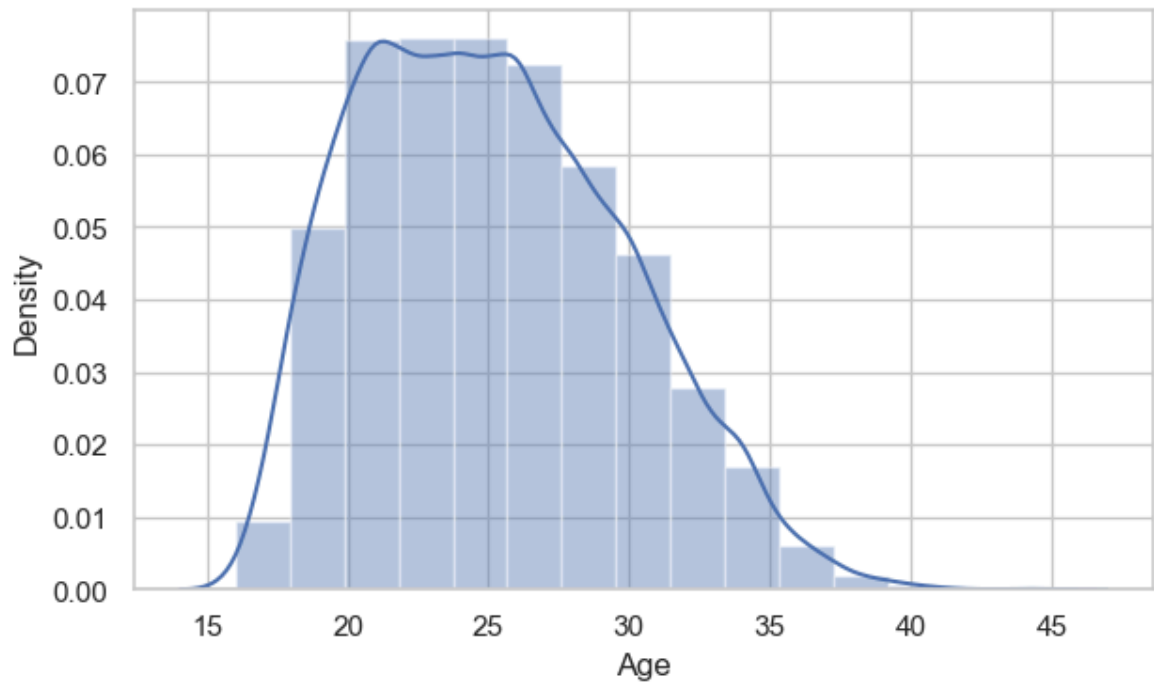
```
In [102...] fifa_data.columns
```

```
Out[102...] Index(['Unnamed: 0', 'ID', 'Name', 'Age', 'Photo', 'Nationality', 'Flag',
      'Overall', 'Potential', 'Club', 'Club Logo', 'Value', 'Wage', 'Special',
      'Preferred Foot', 'International Reputation', 'Weak Foot',
      'Skill Moves', 'Work Rate', 'Body Type', 'Real Face', 'Position',
      'Jersey Number', 'Joined', 'Loaned From', 'Contract Valid Until',
      'Height', 'Weight', 'LS', 'ST', 'RS', 'LW', 'LF', 'CF', 'RF', 'RW',
      'LAM', 'CAM', 'RAM', 'LM', 'LCM', 'CM', 'RCM', 'RM', 'LWB', 'LDM',
      'CDM', 'RDM', 'RWB', 'LB', 'LCB', 'CB', 'RCB', 'RB', 'Crossing',
      'Finishing', 'HeadingAccuracy', 'ShortPassing', 'Volleys', 'Dribbling',
      'Curve', 'FKAccuracy', 'LongPassing', 'BallControl', 'Acceleration',
      'SprintSpeed', 'Agility', 'Reactions', 'Balance', 'ShotPower',
      'Jumping', 'Stamina', 'Strength', 'LongShots', 'Aggression',
      'Interceptions', 'Positioning', 'Vision', 'Penalties', 'Composure',
      'Marking', 'StandingTackle', 'SlidingTackle', 'GKDividing', 'GKHandling',
      'GKKicking', 'GKPositioning', 'GKReflexes', 'Release Clause'],
      dtype='object')
```

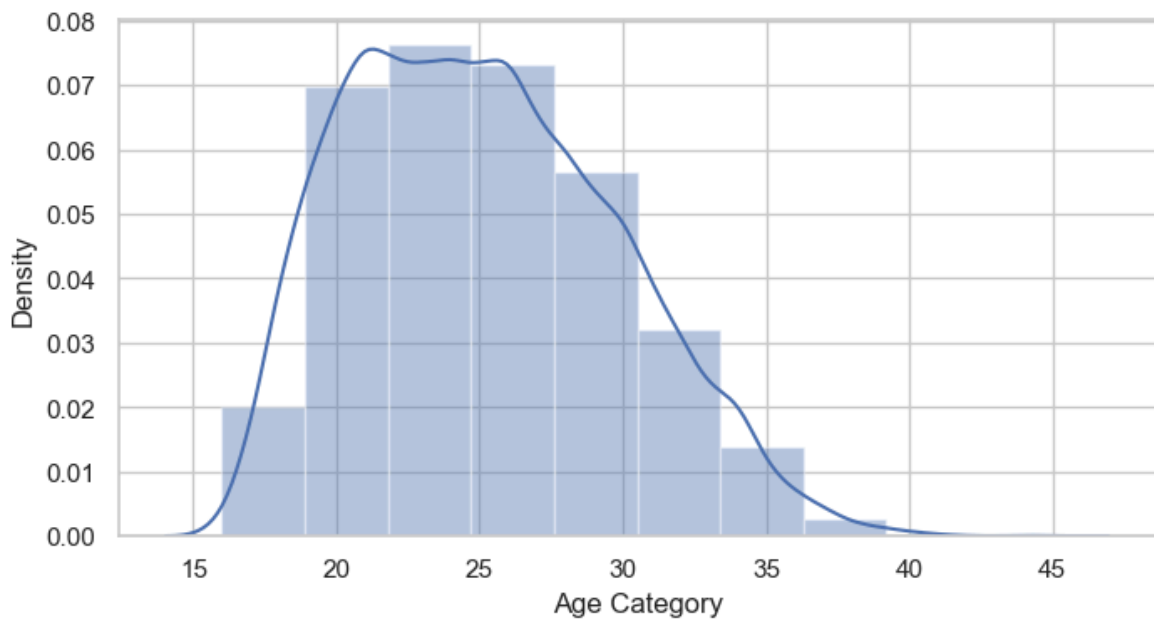
```
In [103...] fifa_data['Body Type'].value_counts()
```

```
Out[103...] Body Type
Normal      10595
Lean        6417
Stocky      1140
Messi        1
C. Ronaldo  1
Neymar       1
Courtois     1
PLAYER_BODY_TYPE_25  1
Shaqiri      1
Akinfenwa    1
Name: count, dtype: int64
```

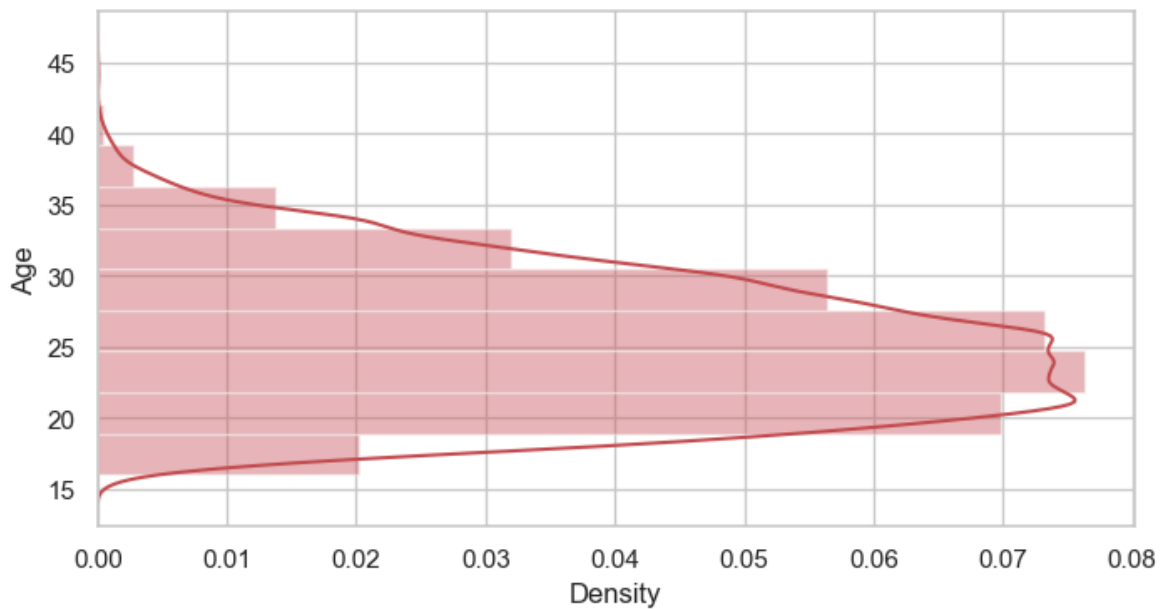
```
In [104...] #vis1 = sns.displot(fifa_data['Age'], bins = 15)
f, ax = plt.subplots(figsize=(7,4))
x = fifa_data['Age']
ax = sns.distplot(x, bins=15)
plt.show()
```



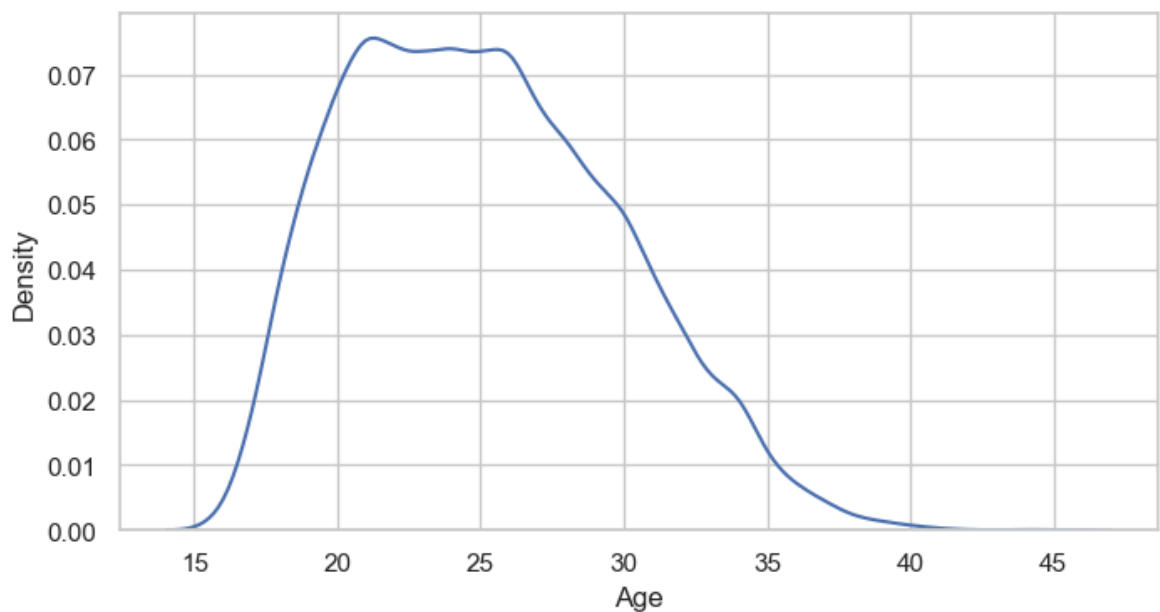
```
In [105... f, ax = plt.subplots(figsize=(8,4))
x = fifa_data['Age']
x = pd.Series(x,name = 'Age Category')
ax = sns.distplot(x, bins = 10)
plt.show()
```



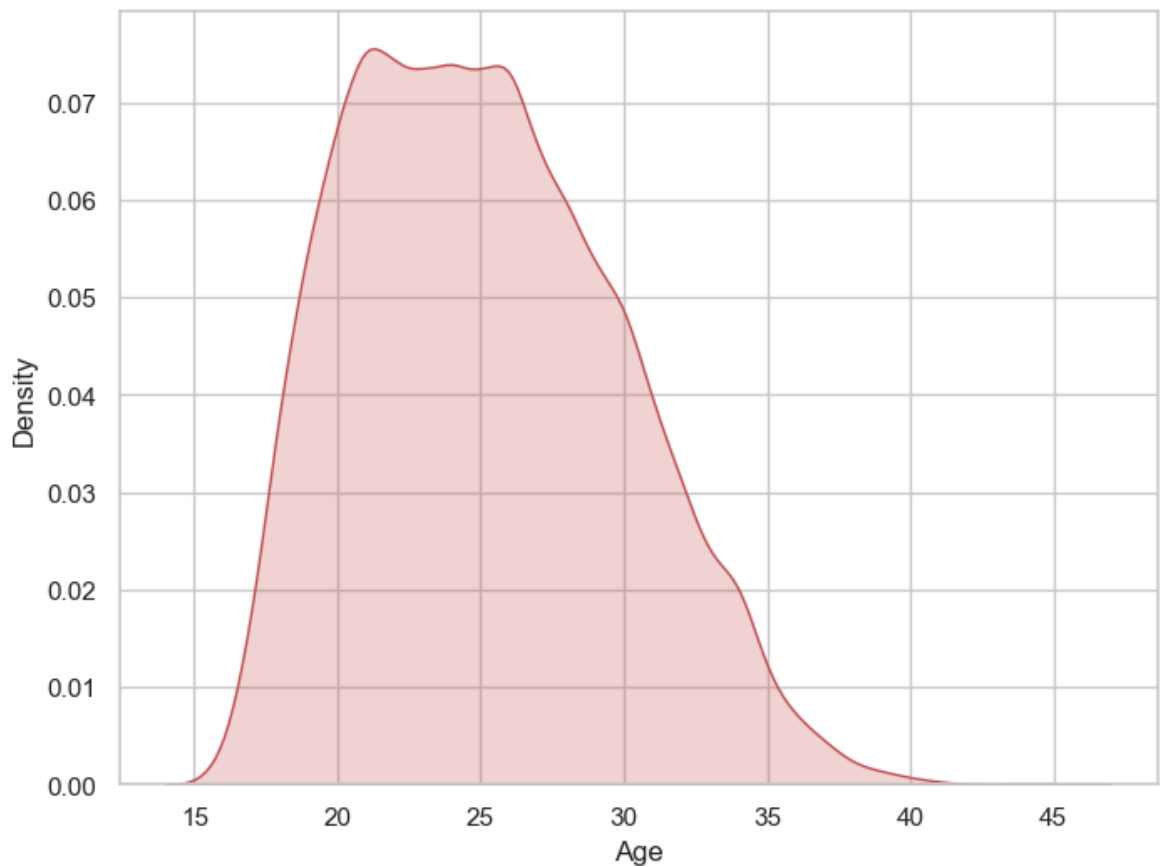
```
In [106... f, ax = plt.subplots(figsize = (8,4))
x = fifa_data['Age']
ax = sns.distplot(x, bins = 10, vertical = True, color = 'r')
plt.show()
```



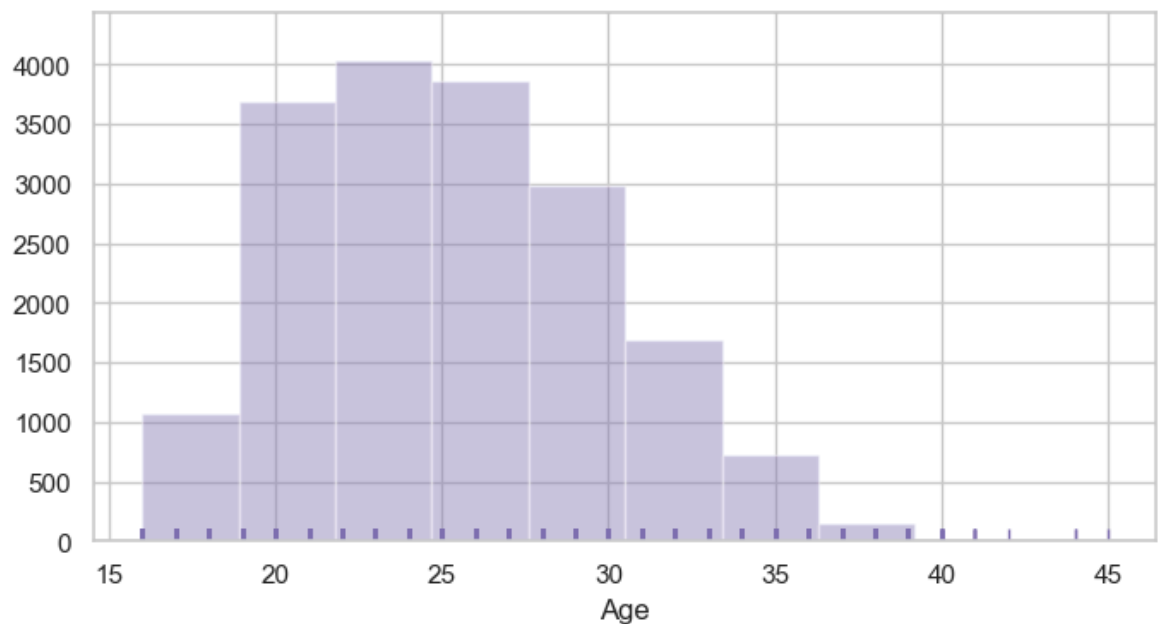
```
In [107... f, ax = plt.subplots(figsize=(8, 4))
x = fifa_data['Age']
ax = sns.kdeplot(x, shade=False)
plt.show()
```



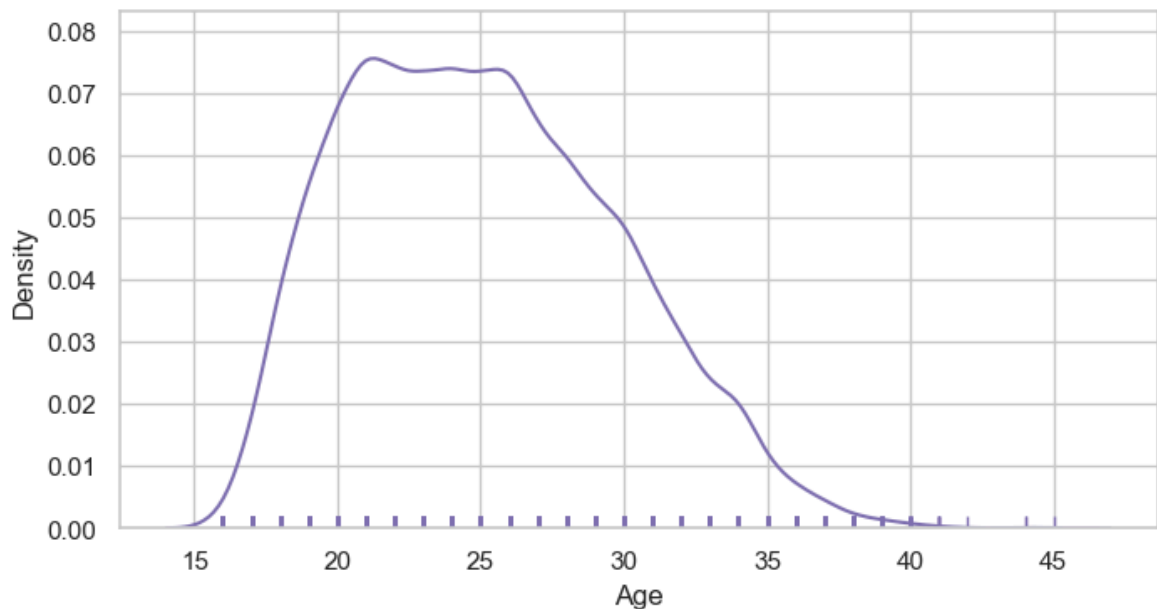
```
In [108... f, ax = plt.subplots(figsize=(8, 6))
x = fifa_data['Age']
ax = sns.kdeplot(x, color="r", shade=True) #shade=True will fill the area under
plt.show()
```



```
In [109... f, ax = plt.subplots(figsize=(8, 4))
x = fifa_data['Age']
ax = sns.distplot(x, kde=False, color='m', rug=True, bins = 10) #rug will show
plt.show()
```



```
In [110... f, ax = plt.subplots(figsize = (8, 4))
x = fifa_data['Age']
ax = sns.distplot(x, hist = False, kde = True, rug = True, color = 'm')#when his
plt.show()
```



```
In [111...] fifa_data['Preferred Foot'].value_counts()
```

```
Out[111...] Preferred Foot
Right      13948
Left       4211
Name: count, dtype: int64
```

```
In [112...] fifa_data['Preferred Foot'].unique()#unique() will return all the unique values
```

```
Out[112...] array(['Left', 'Right', nan], dtype=object)
```

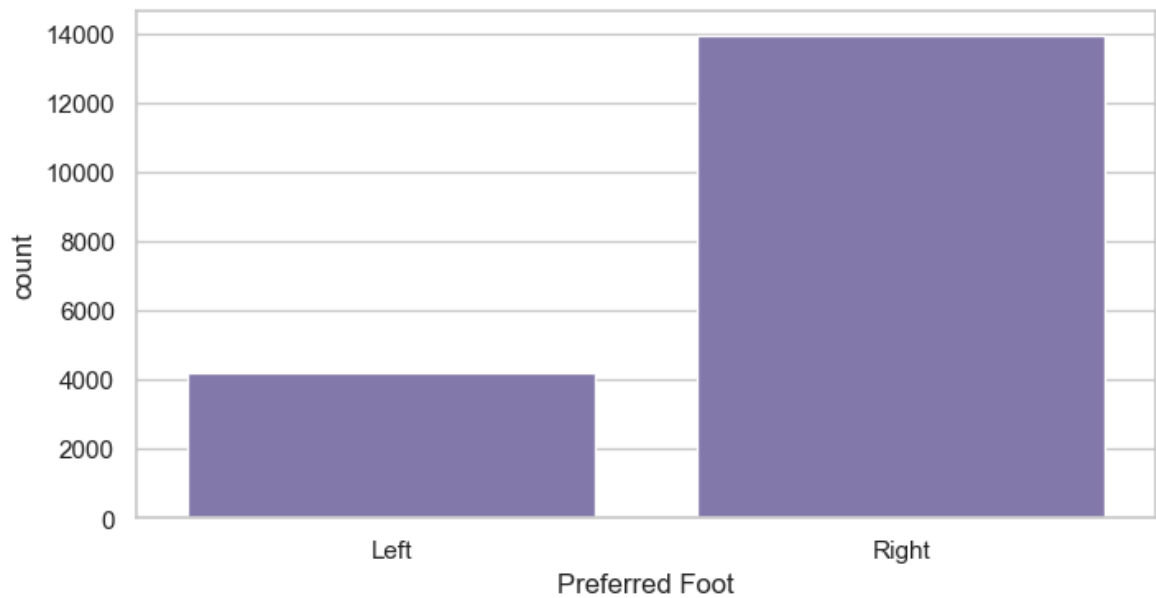
```
In [113...] fifa_data['Preferred Foot'].nunique()#nunique() will give the number of unique v
```

```
Out[113...] 2
```

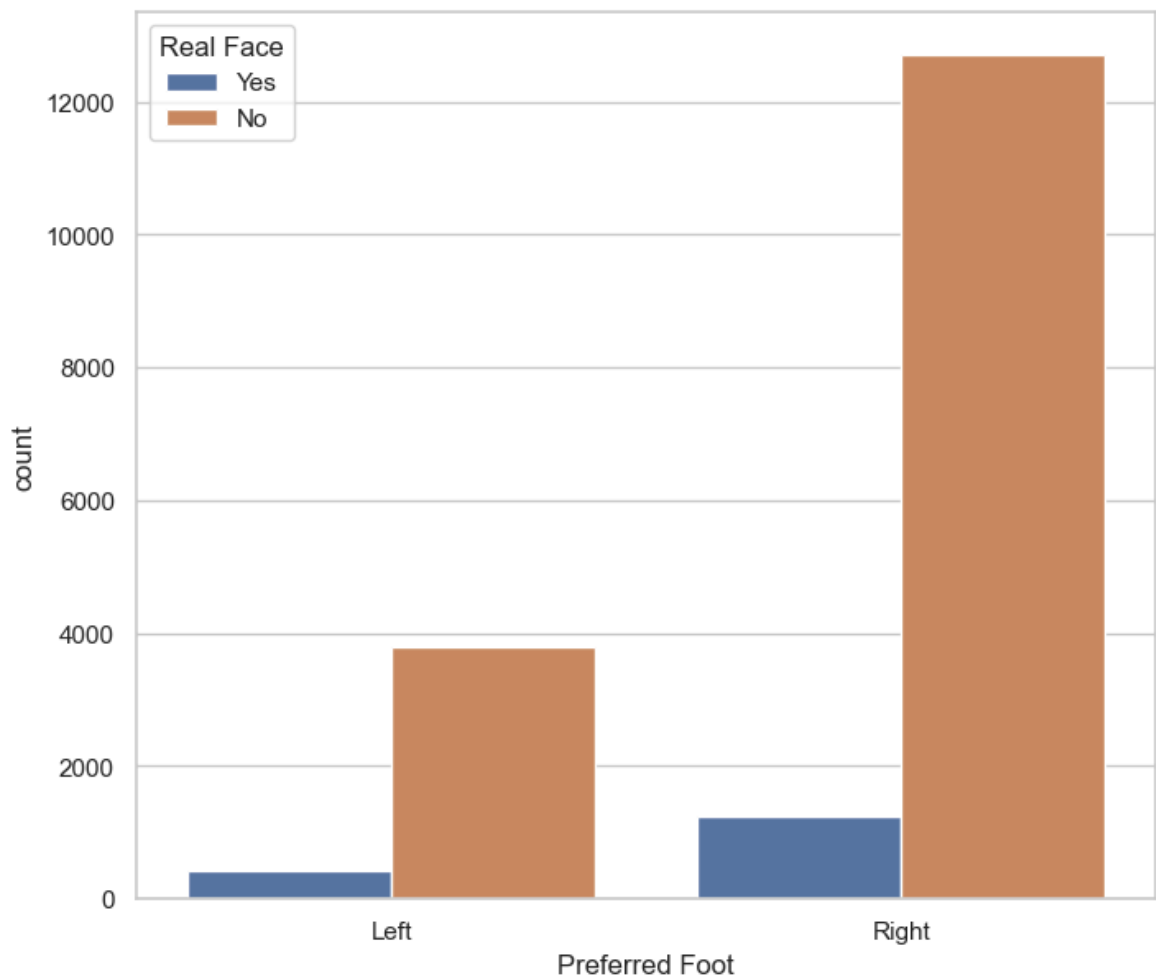
```
In [114...] fifa_data['Preferred Foot'].value_counts()
```

```
Out[114...] Preferred Foot
Right      13948
Left       4211
Name: count, dtype: int64
```

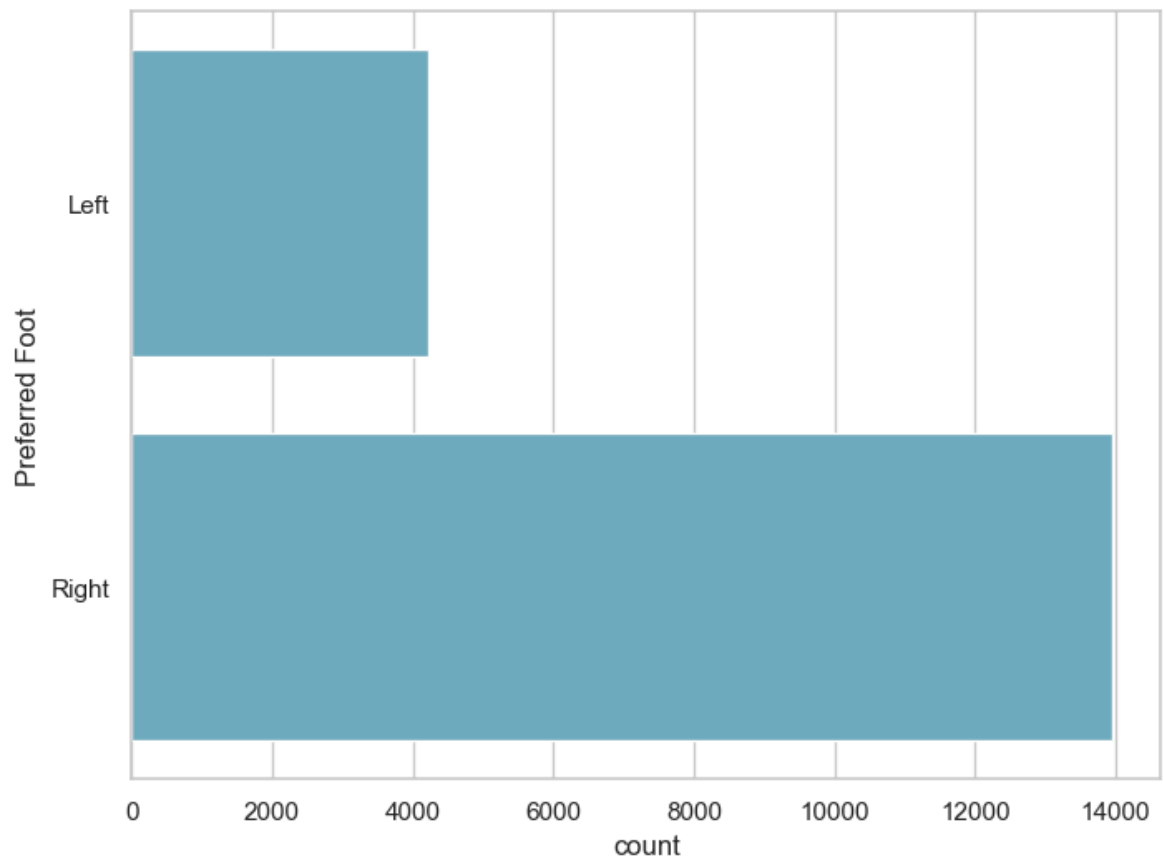
```
In [115...] f, ax = plt.subplots(figsize=(8,4))
sns.countplot(x='Preferred Foot', data = fifa_data, color = 'm')
plt.show()
```



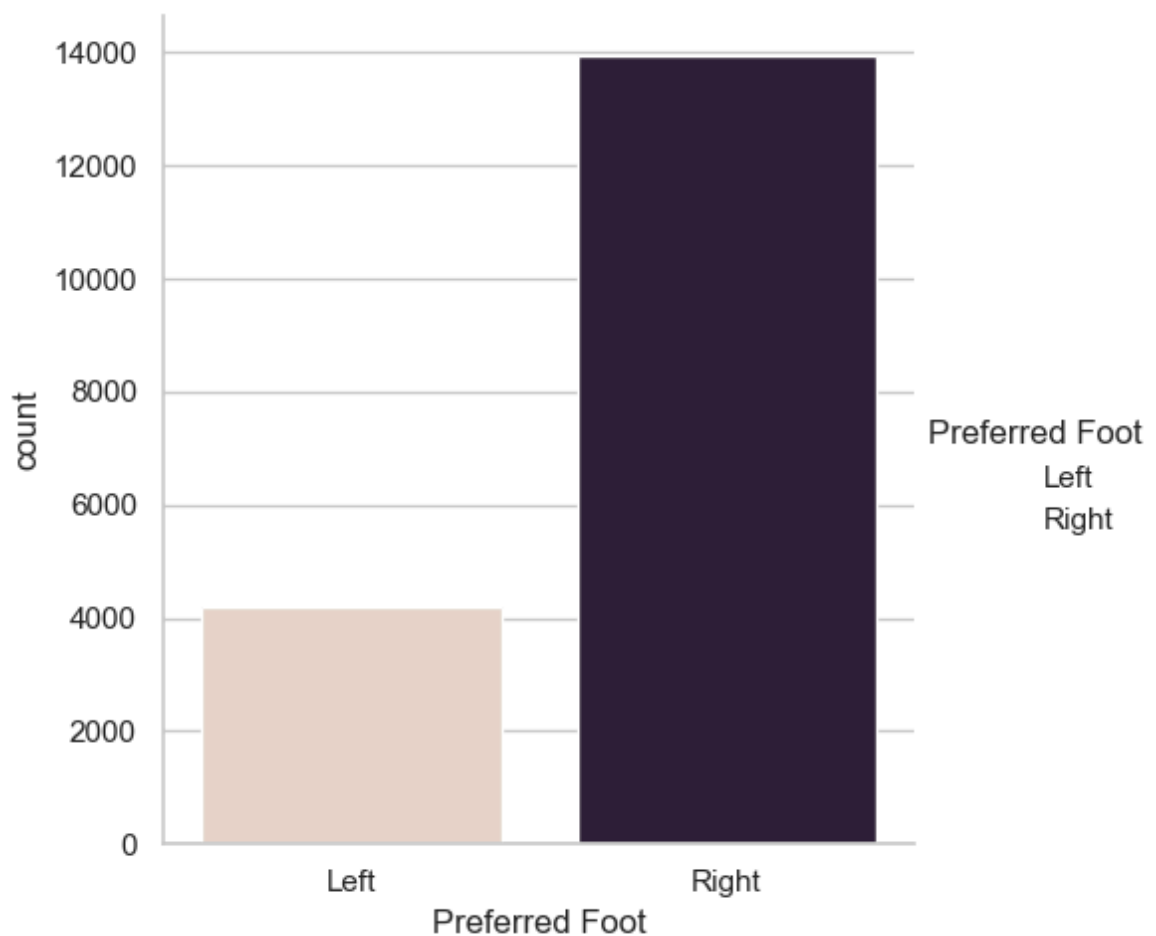
```
In [116... f, ax = plt.subplots(figsize=(8, 7))
sns.countplot(x = 'Preferred Foot', hue = 'Real Face', data = fifa_data)#hue = '
plt.show()
```



```
In [117... f, ax = plt.subplots(figsize=(8,6))
sns.countplot(y = 'Preferred Foot', data=fifa_data, color='c')
plt.show()
```



```
In [118...] vis2 = sns.catplot(x = 'Preferred Foot', data=fifa_data, kind='count', palette =
```



```
In [119...] fifa_data['International Reputation'].nunique()
```

Out[119...] 5

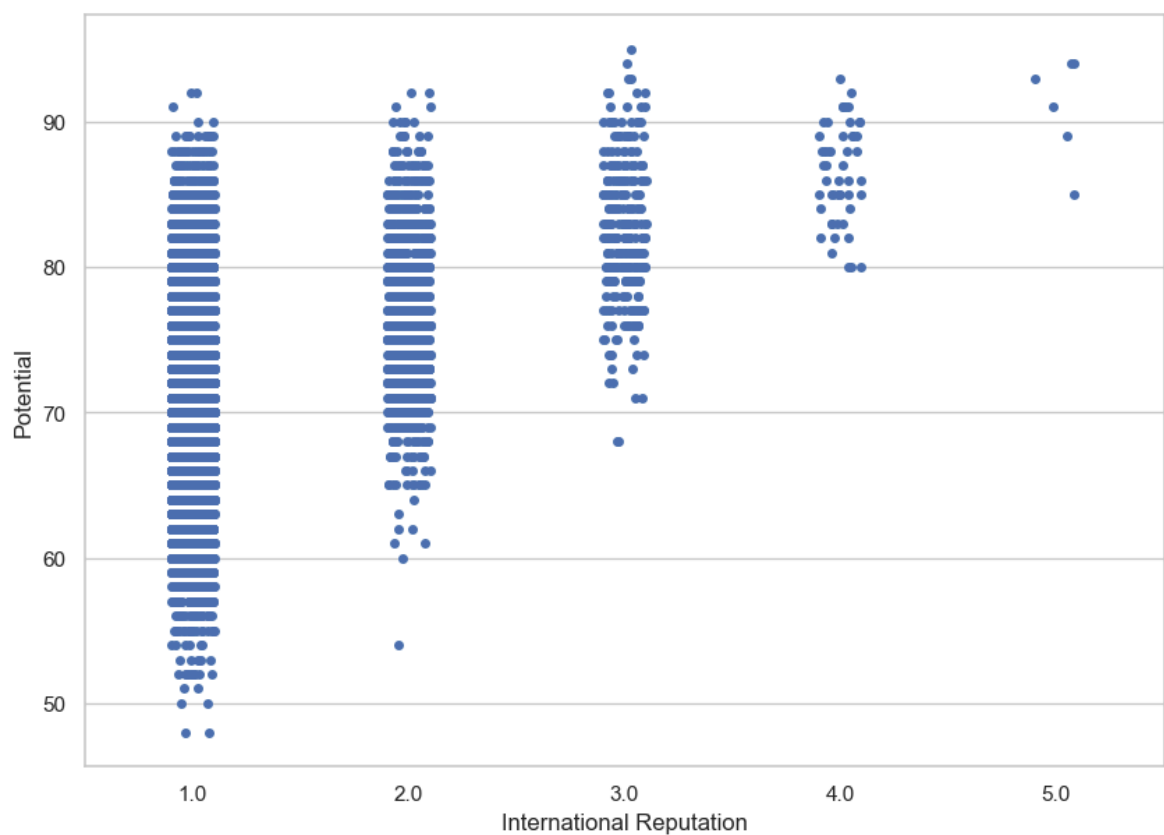
In [120...] `fifa_data['International Reputation'].value_counts()`

Out[120...] International Reputation

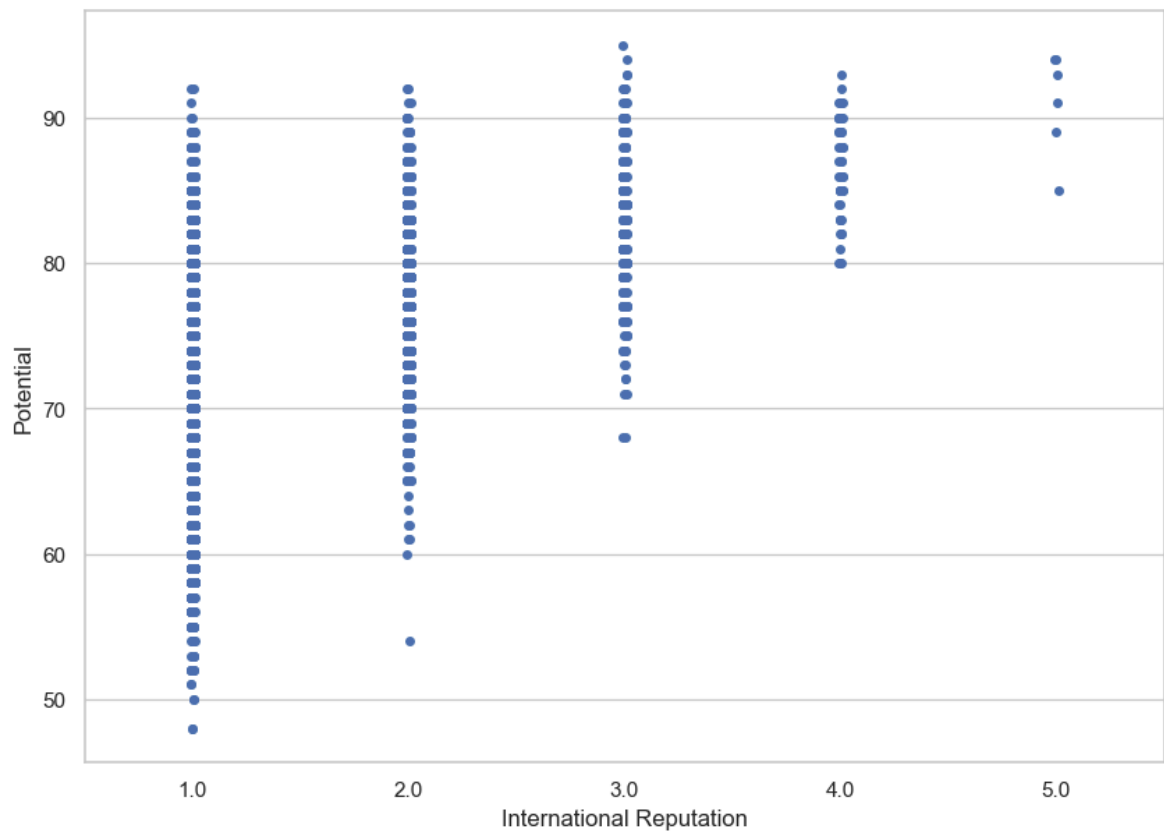
1.0	16532
2.0	1261
3.0	309
4.0	51
5.0	6

Name: count, dtype: int64

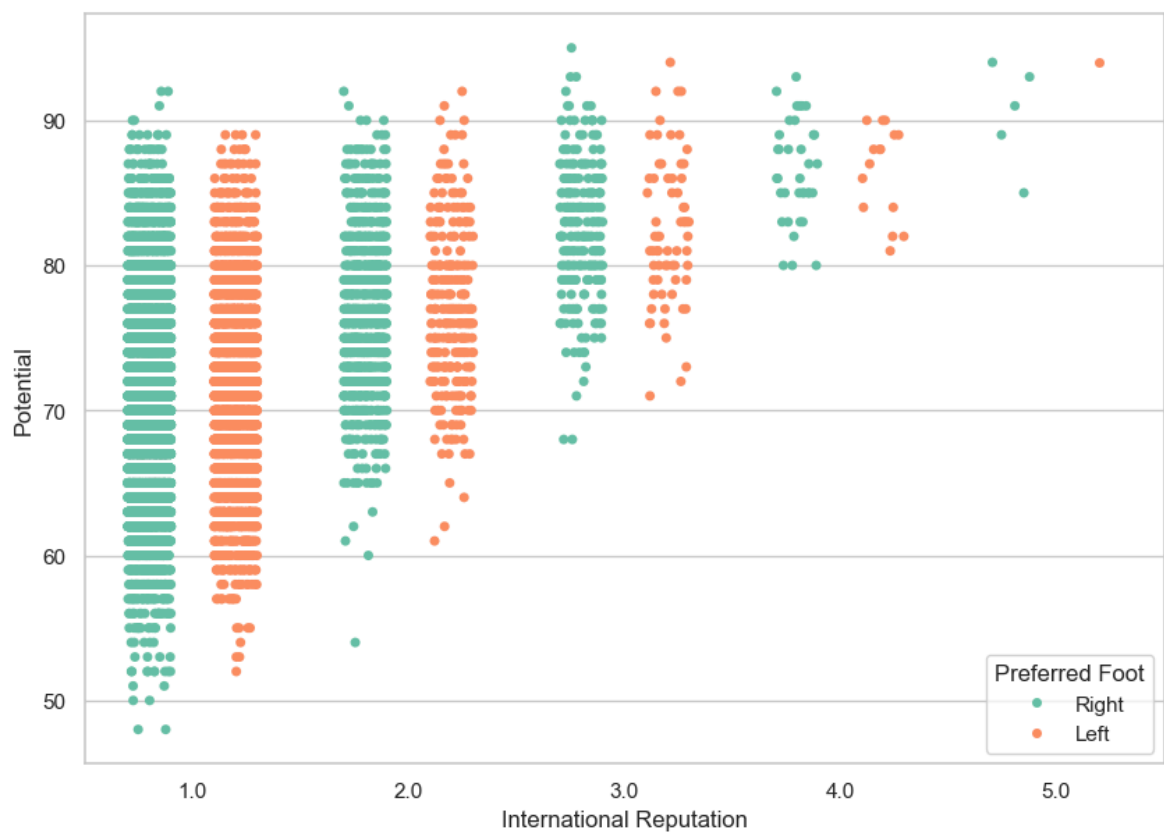
In [121...] `f, ax = plt.subplots(figsize=(10,7))`
`sns.stripplot(x = 'International Reputation', y = 'Potential', data = fifa_data)`
`plt.show`

Out[121...] `<function matplotlib.pyplot.show(close=None, block=None)>`

In [122...] `f, ax = plt.subplots(figsize=(10,7))`
`sns.stripplot(x = 'International Reputation', y = 'Potential', data = fifa_data,`
`plt.show())`

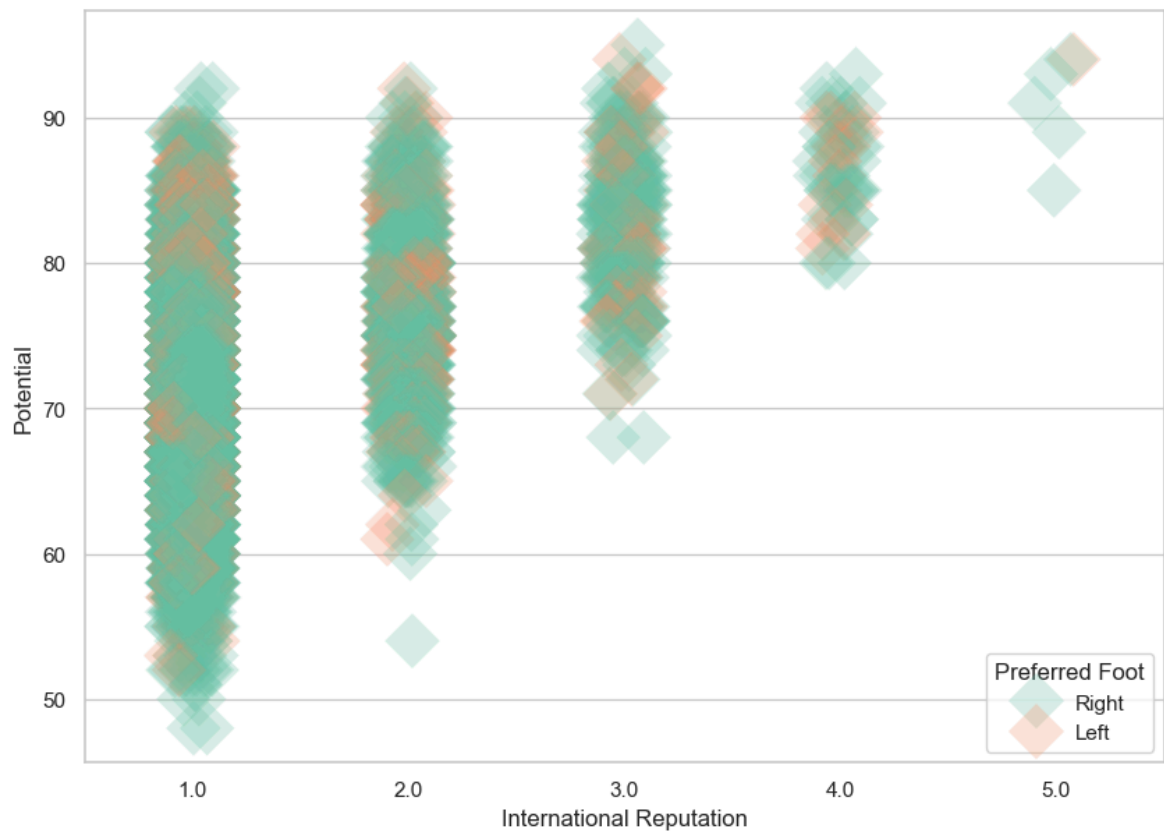


```
In [123... f, ax = plt.subplots(figsize=(10, 7))
sns.stripplot(x='International Reputation', y = 'Potential', hue='Preferred Foot'
plt.show()
```



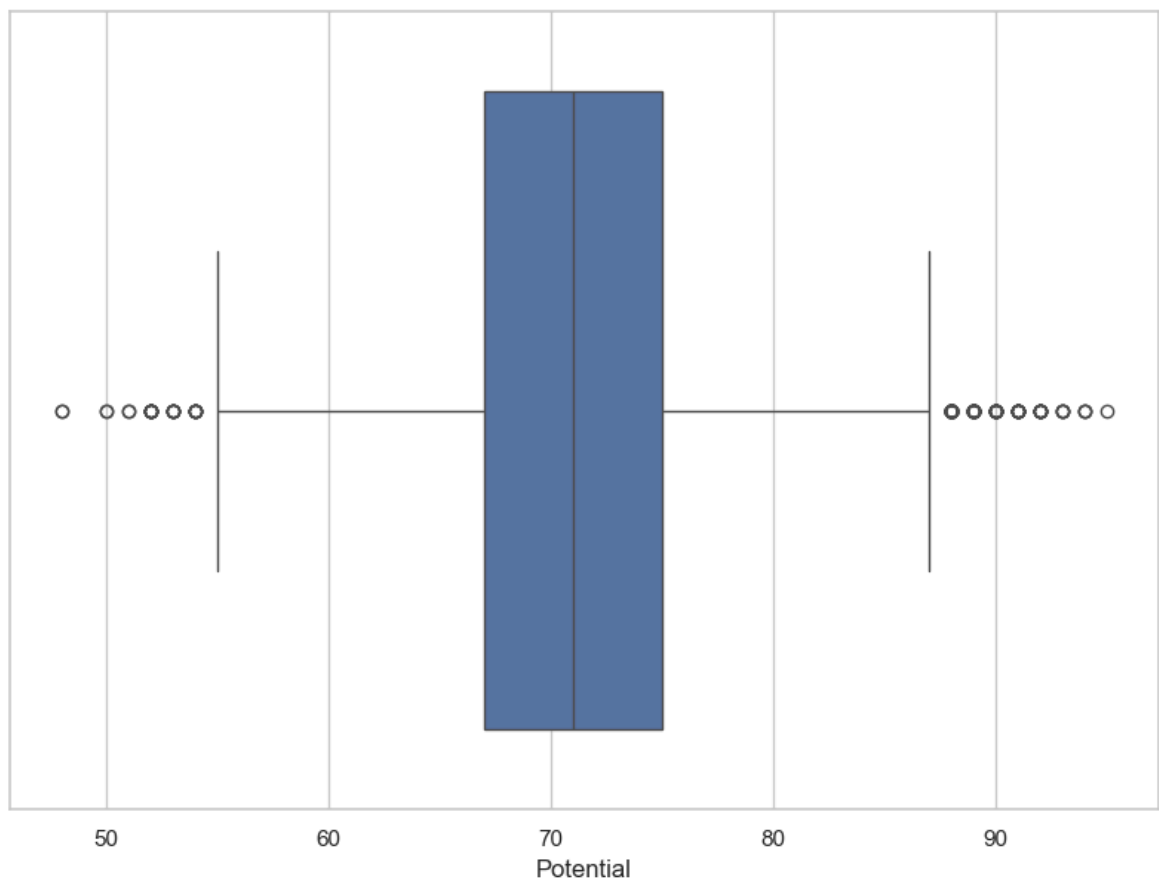
```
In [124... f, ax = plt.subplots(figsize=(10, 7))
sns.stripplot(x='International Reputation', y='Potential', data=fifa_data, hue =
```

```
Out[124... <Axes: xlabel='International Reputation', ylabel='Potential'>
```

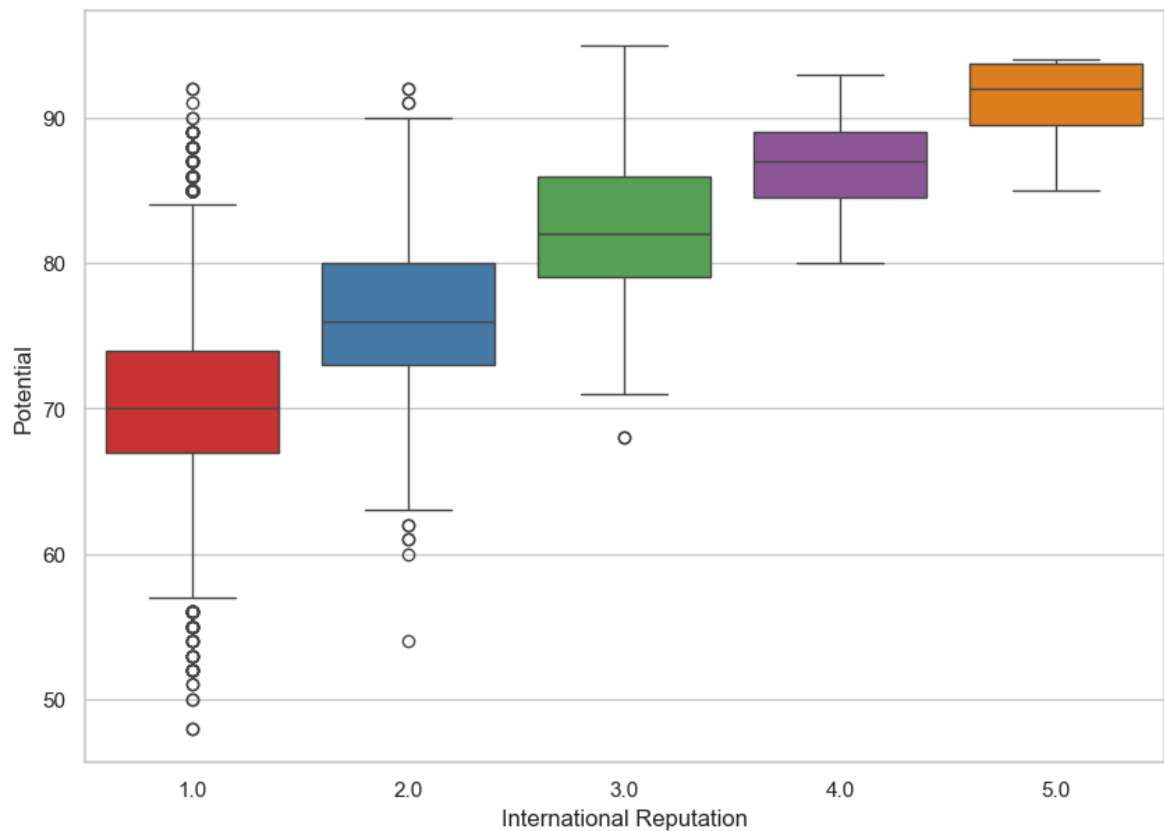


In [125...

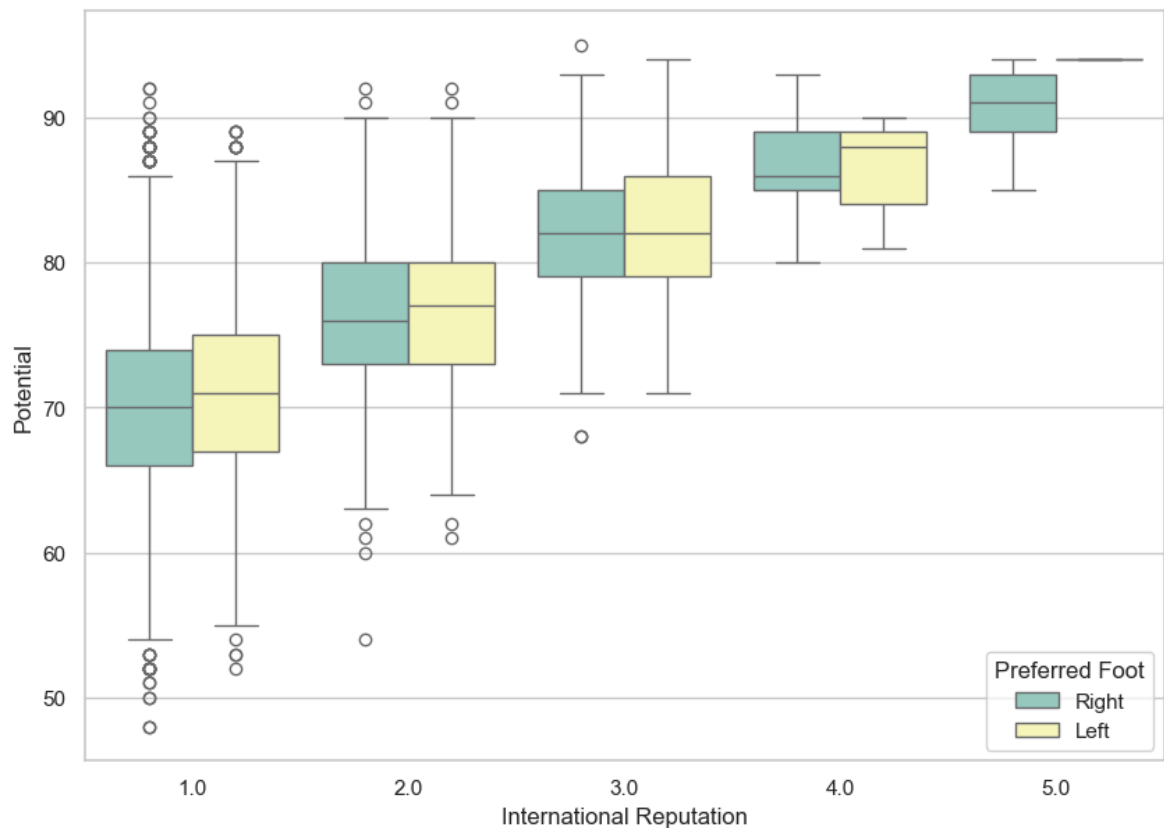
```
f, ax = plt.subplots(figsize=(10,7))
sns.boxplot(x=fifa_data['Potential'])#boxplot is a visual representation of the
plt.show()
```



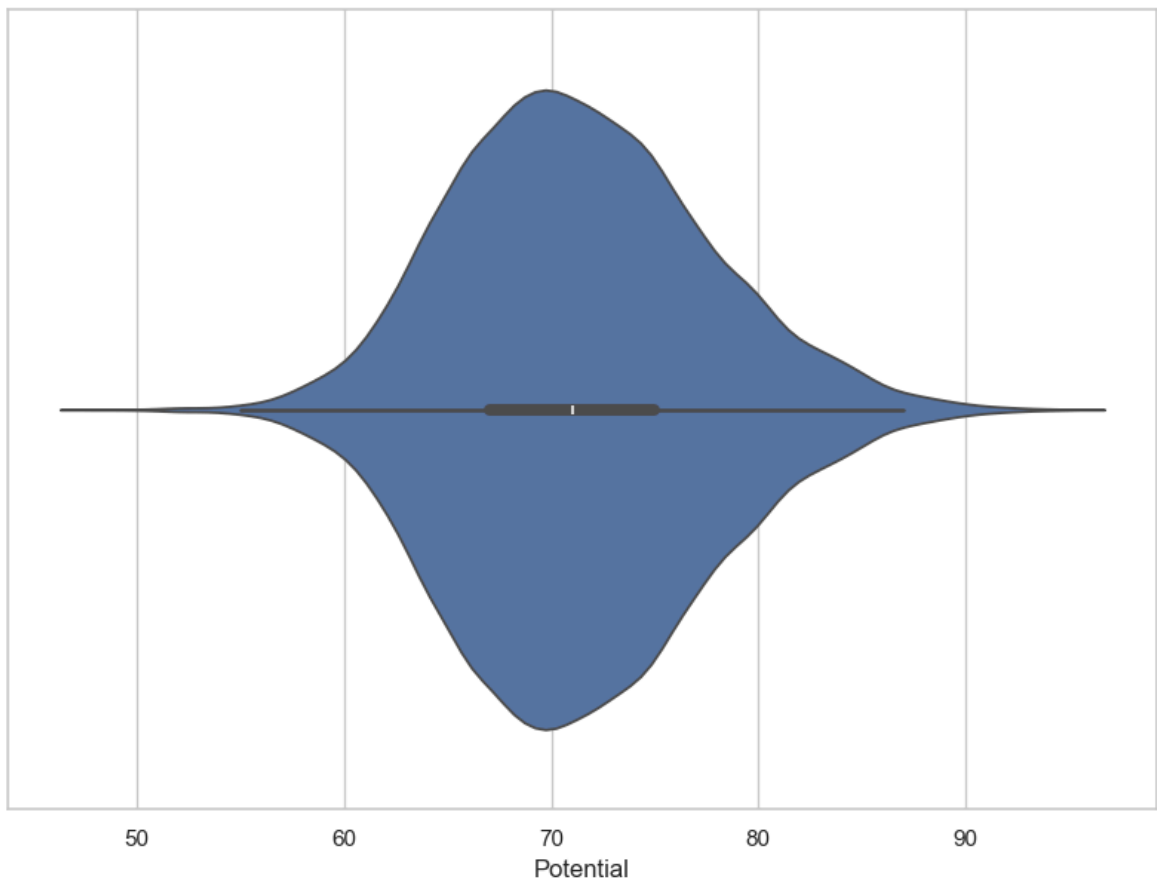
```
In [126... f, ax=plt.subplots(figsize=(10,7))
sns.boxplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show()
```



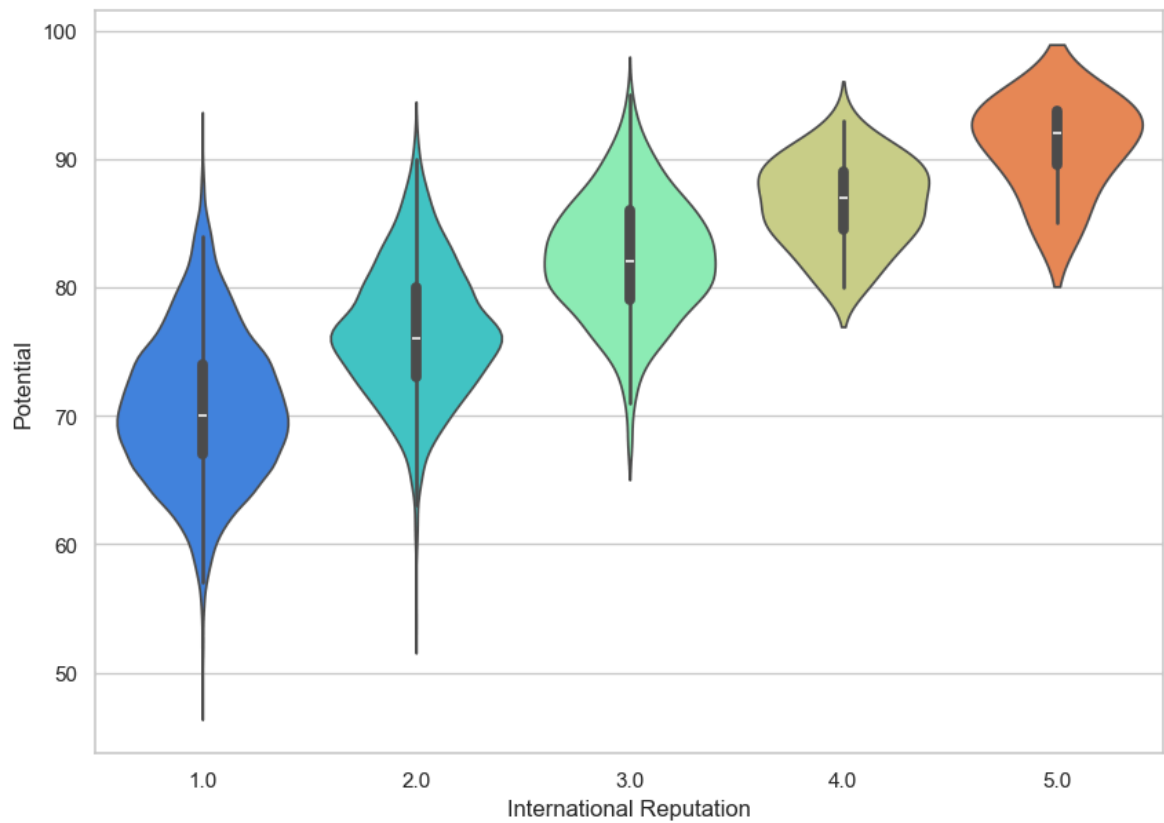
```
In [127... f, ax=plt.subplots(figsize=(10,7))
sns.boxplot(x='International Reputation', y='Potential', data=fifa_data, hue='Pr
plt.show()
```



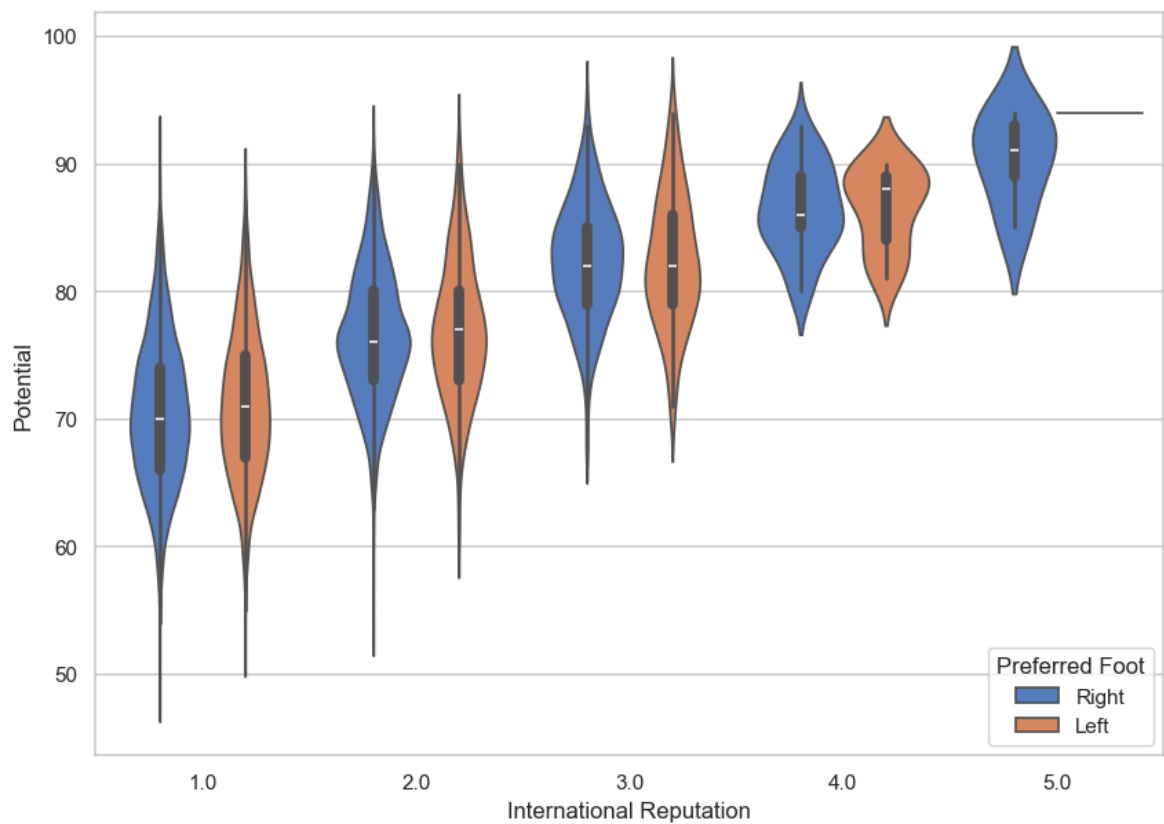
```
In [128... f,ax=plt.subplots(figsize=(10,7))
sns.violinplot(x='Potential', data=fifa_data)#violinplot is used to plot the dis
plt.show()
```



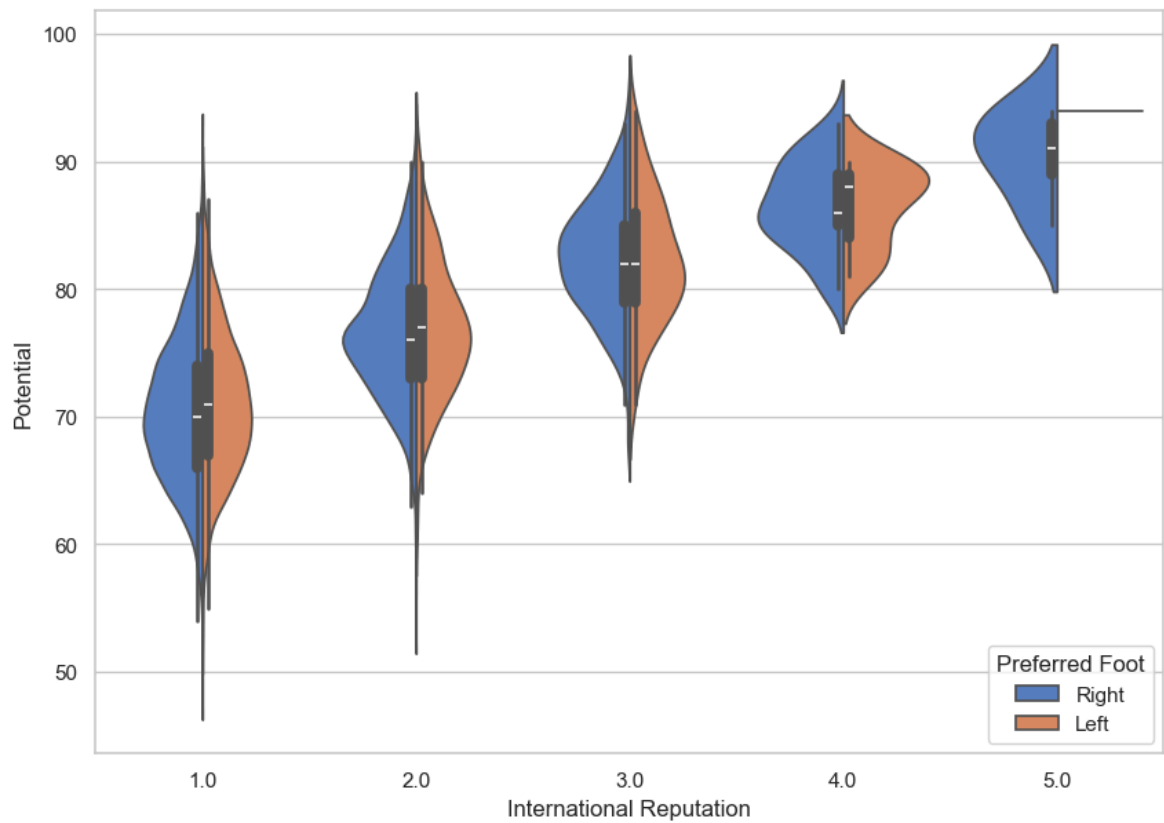
```
In [129... f,ax=plt.subplots(figsize=(10,7))
sns.violinplot(x='International Reputation', y='Potential', data=fifa_data, pale
plt.show())
```



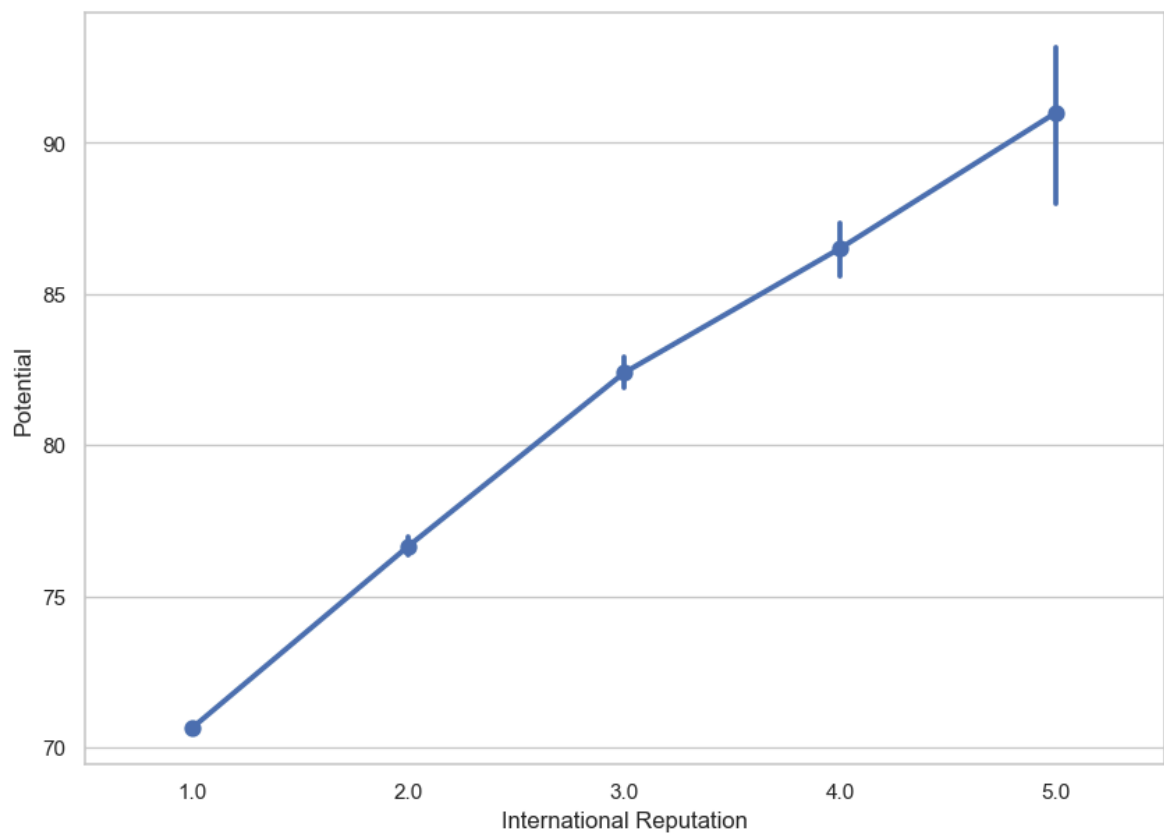
```
In [130... f,ax=plt.subplots(figsize=(10,7))
sns.violinplot(x='International Reputation', y='Potential', data=fifa_data, pale
plt.show()
```



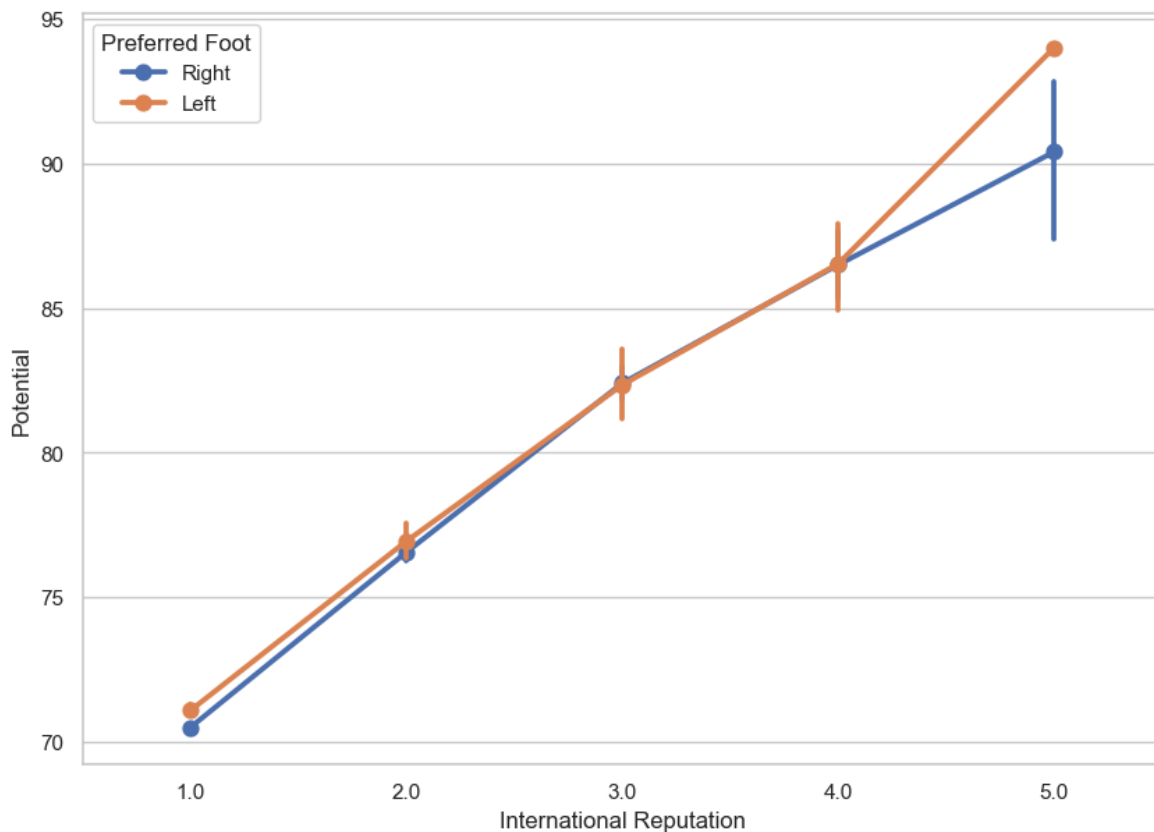
```
In [131... f,ax=plt.subplots(figsize=(10,7))
sns.violinplot(x='International Reputation', y='Potential', data=fifa_data, pale
plt.show()
```



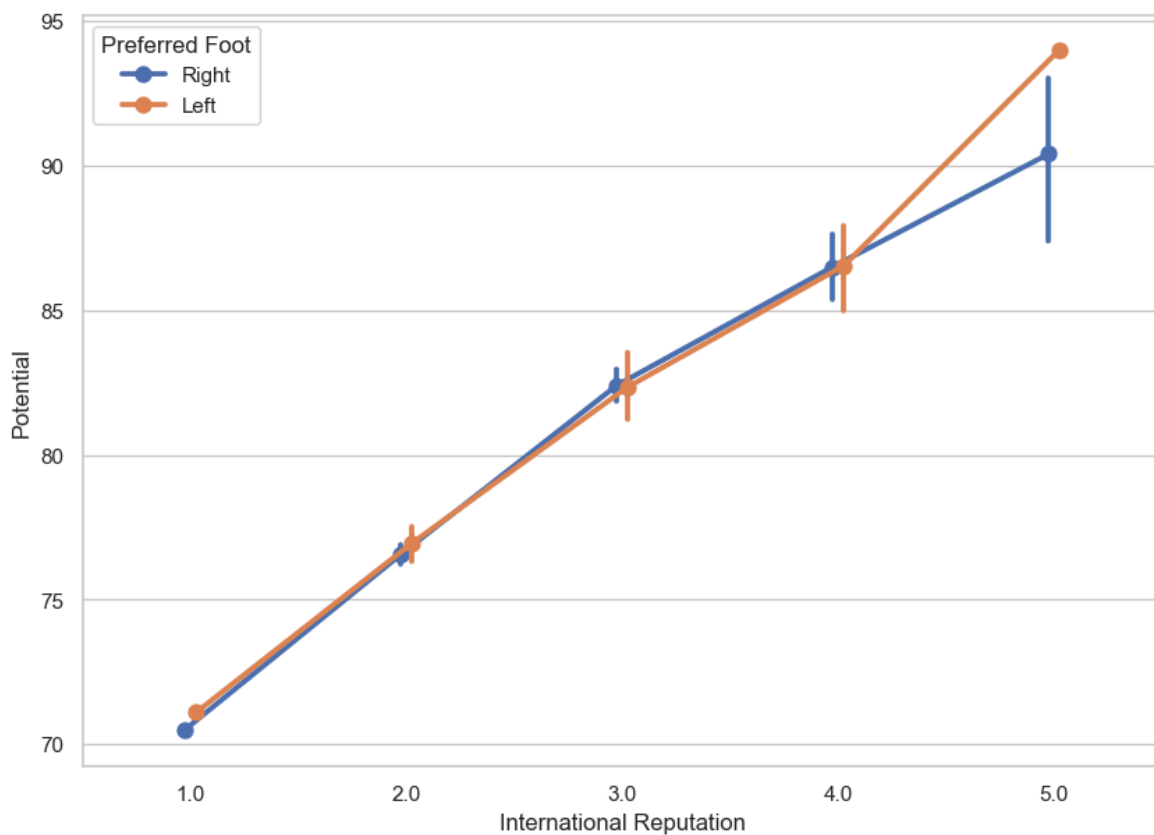
```
In [132... f,ax=plt.subplots(figsize=(10,7))
sns.pointplot(x='International Reputation', y='Potential', data=fifa_data)#point
plt.show()
```



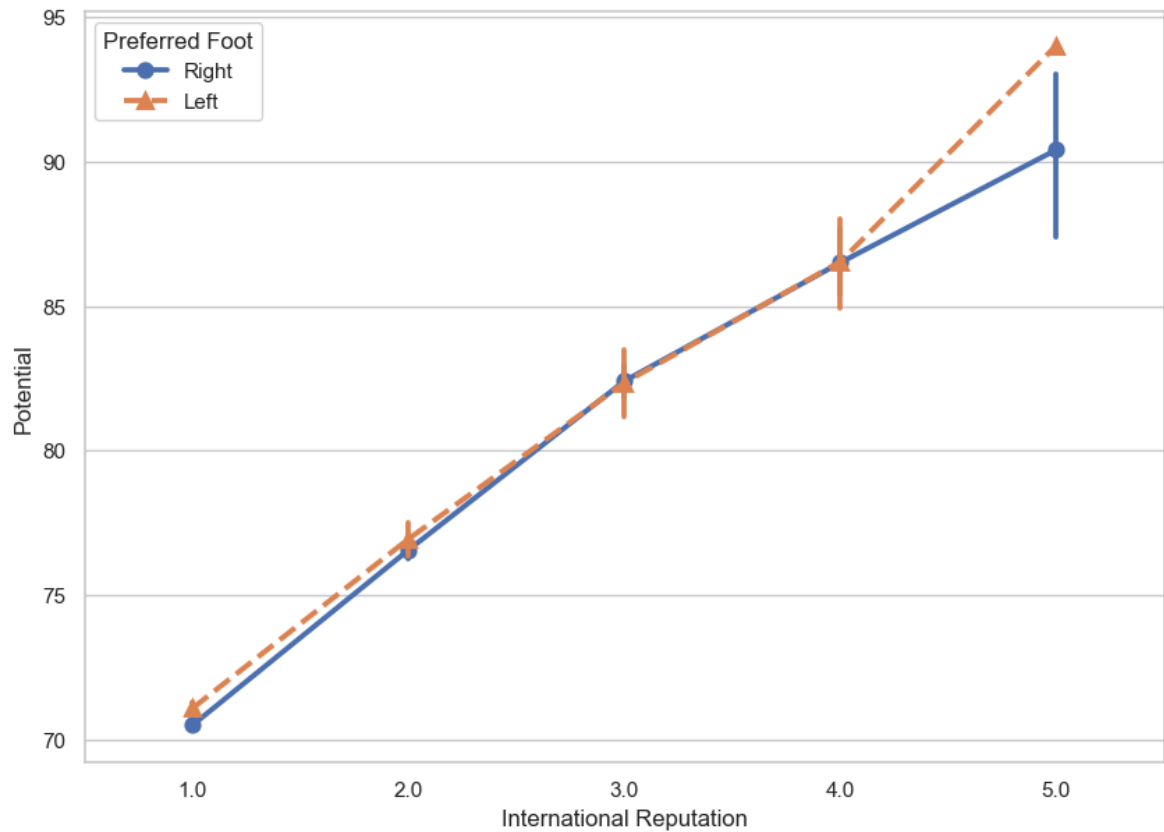
```
In [133... f,ax=plt.subplots(figsize=(10,7))
sns.pointplot(x='International Reputation', y='Potential', data=fifa_data, hue='
plt.show()
```



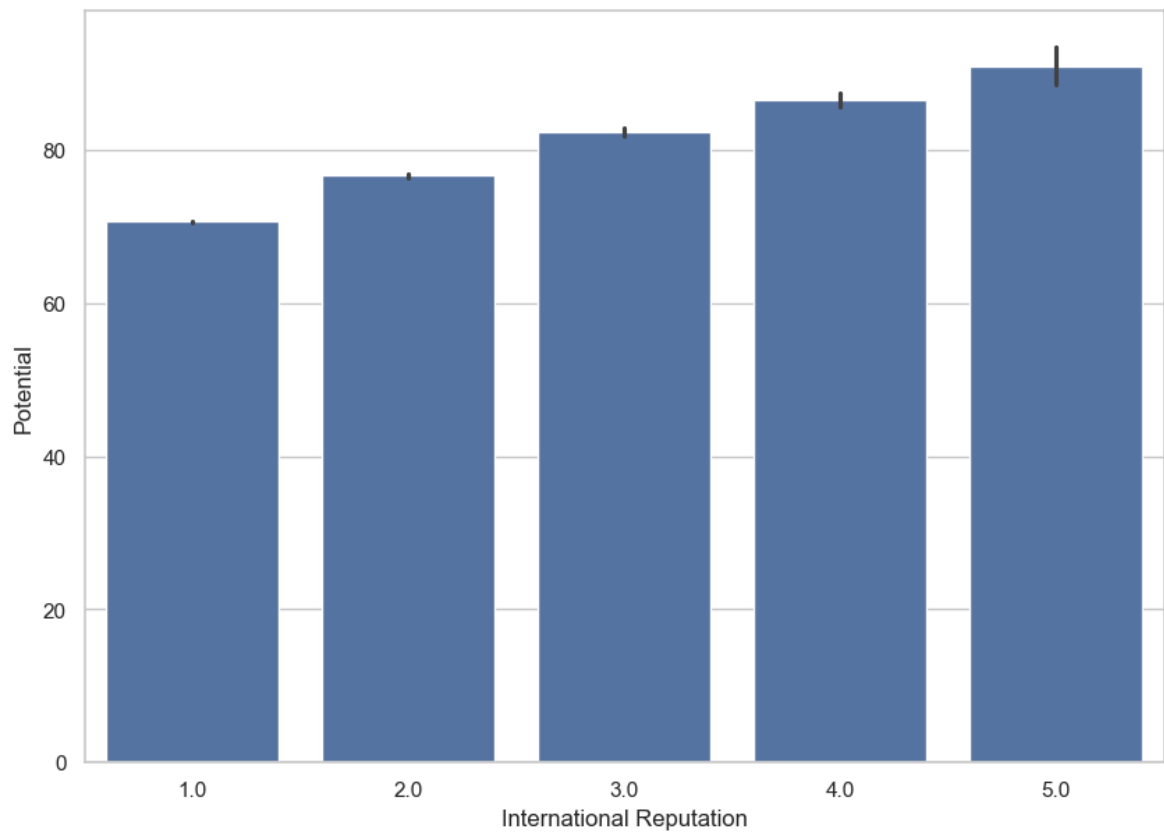
```
In [134... f,ax=plt.subplots(figsize=(10,7))
sns.pointplot(x='International Reputation', y='Potential', data=fifa_data, hue='
plt.show()
```



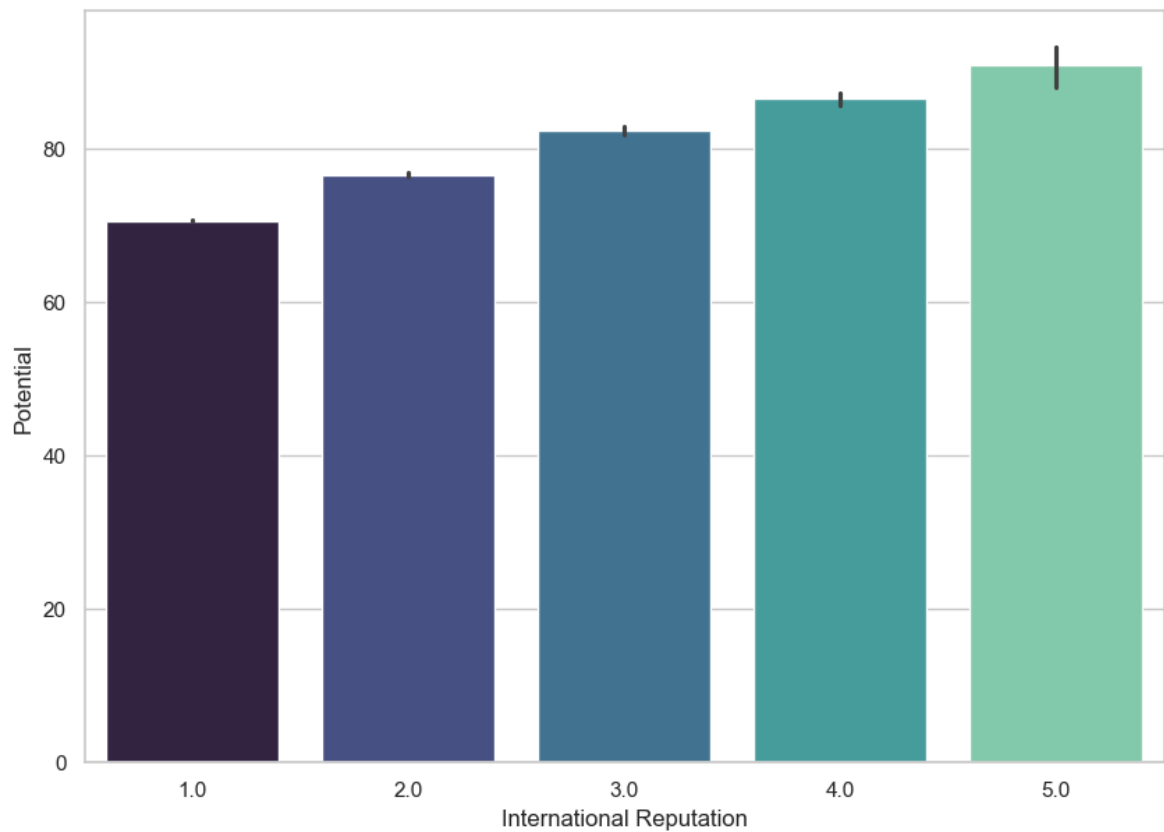
```
In [135... f,ax=plt.subplots(figsize=(10,7))
sns.pointplot(x='International Reputation', y='Potential', data=fifa_data, hue='
plt.show()
```



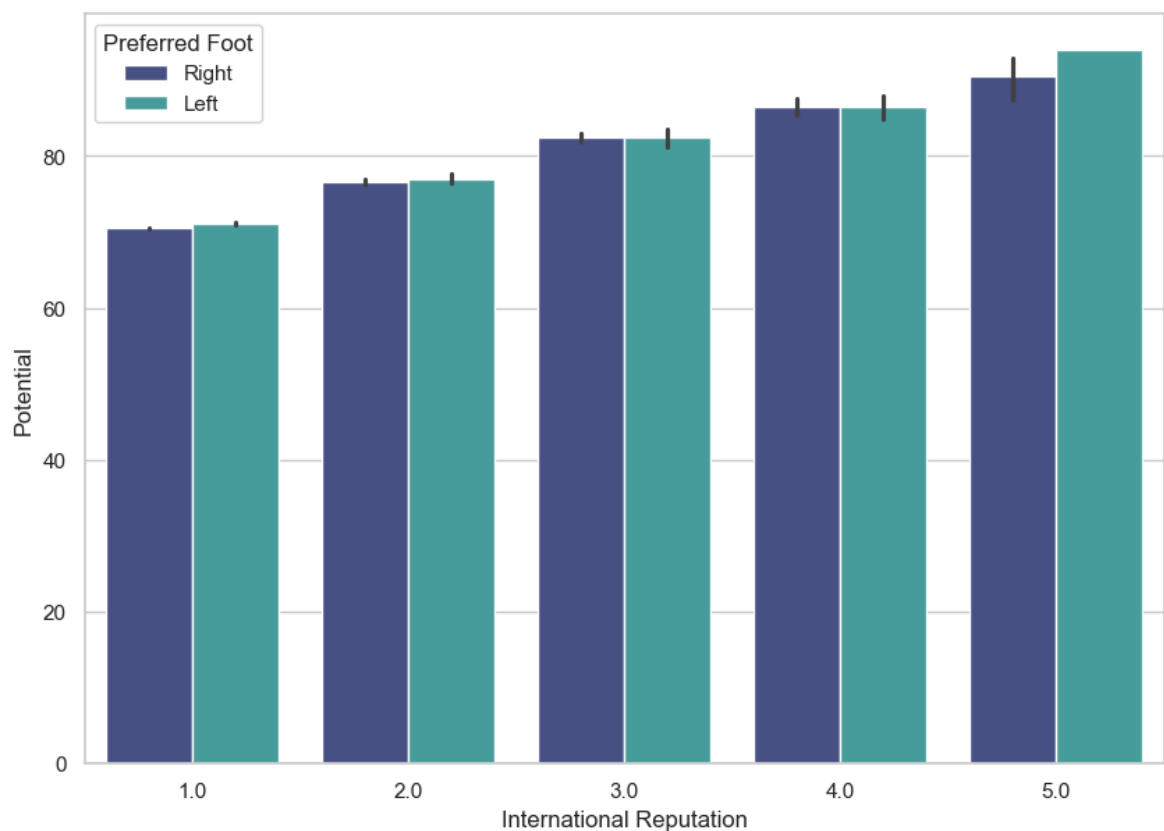
```
In [136... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data)
plt.show()
```



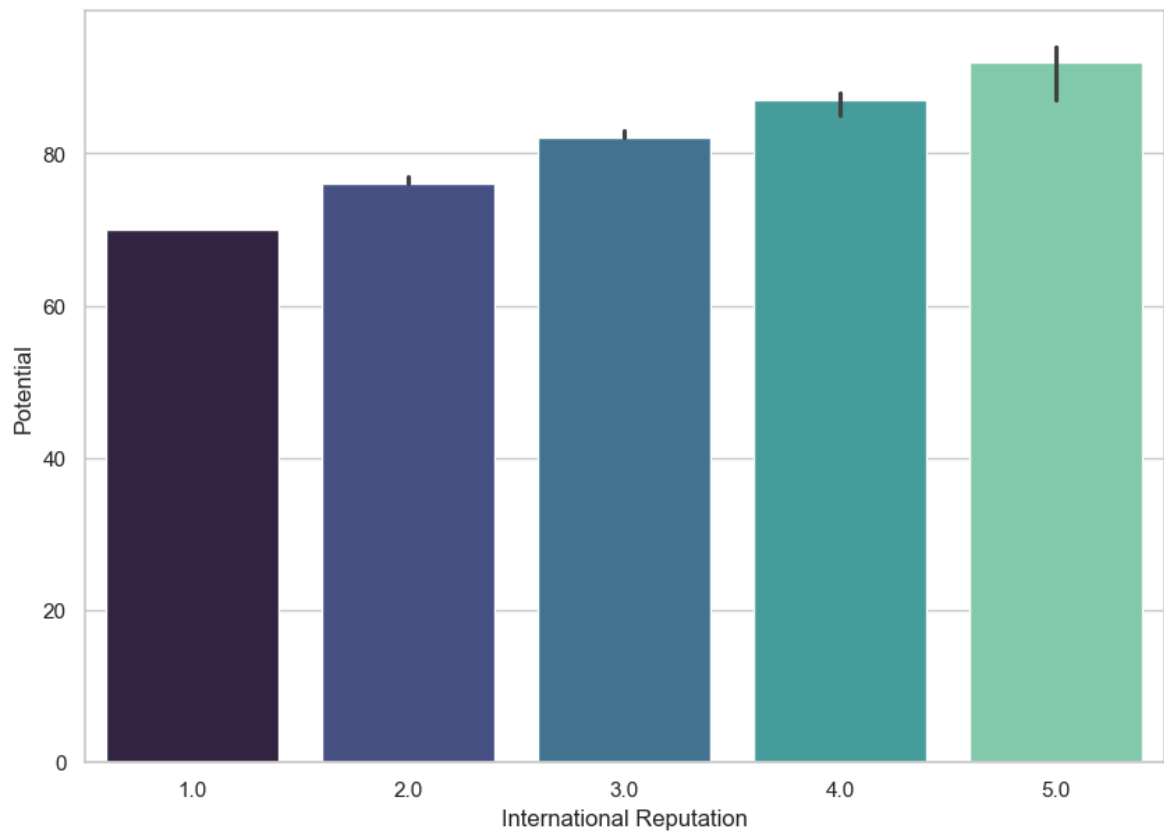
```
In [137... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show()
```



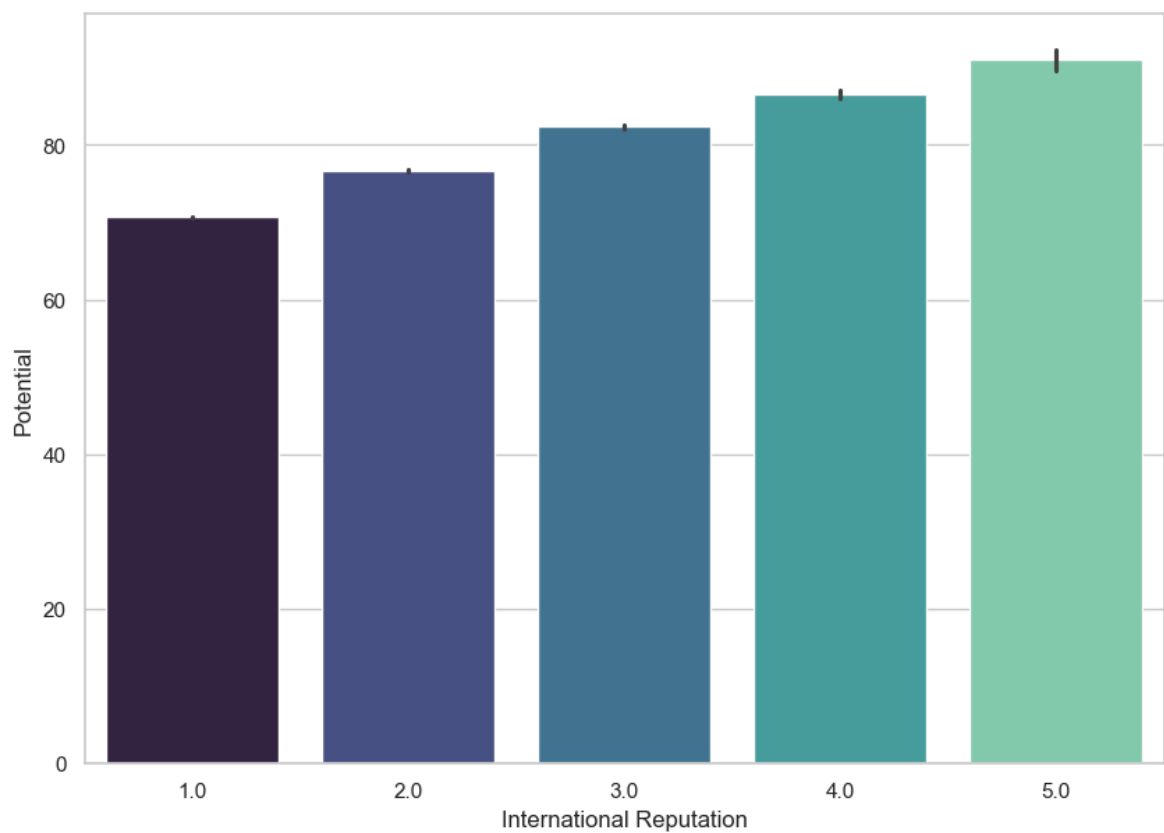
```
In [138... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show())
```



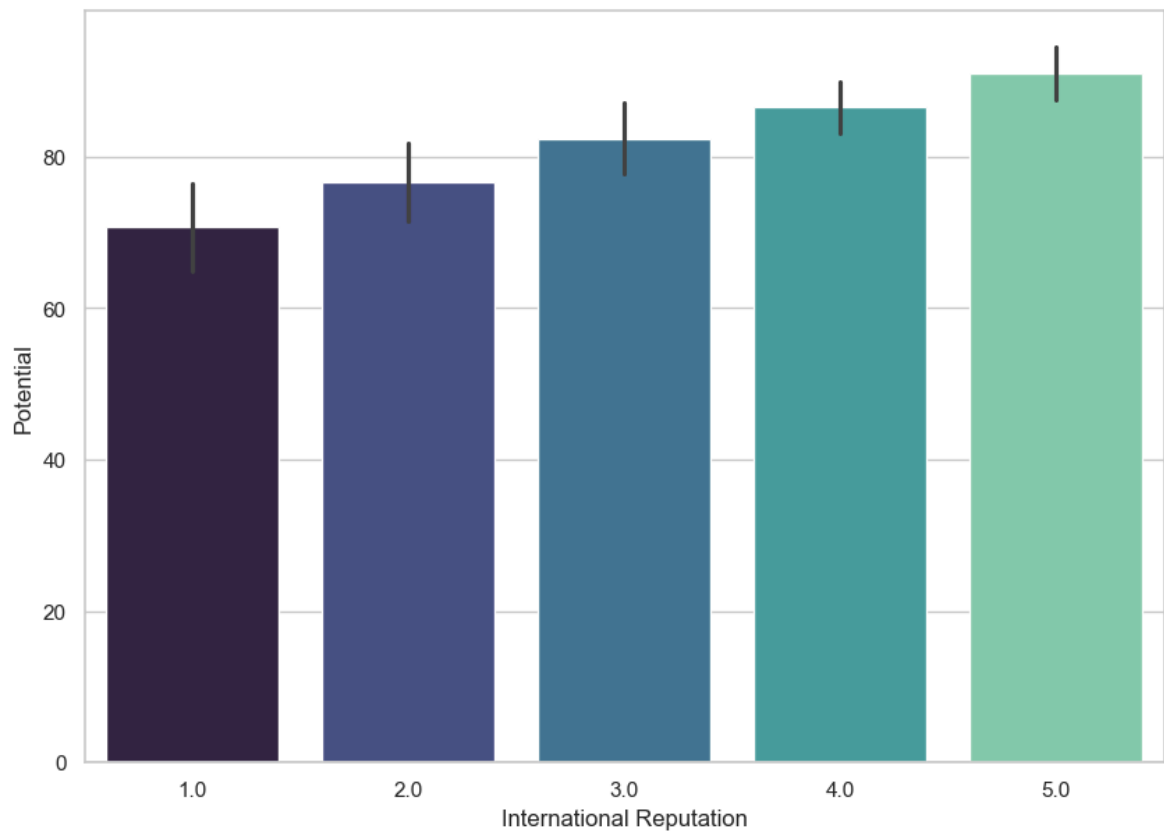
```
In [139... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show())
```



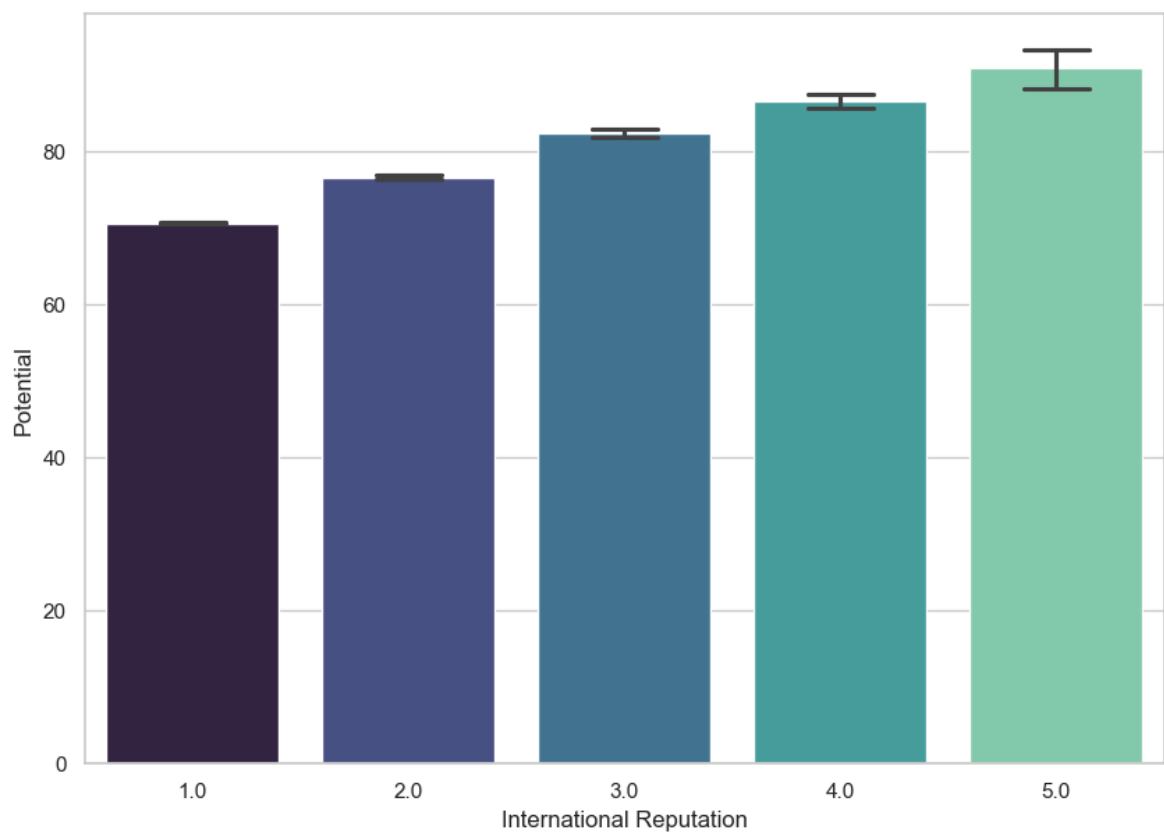
```
In [140... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show())
```



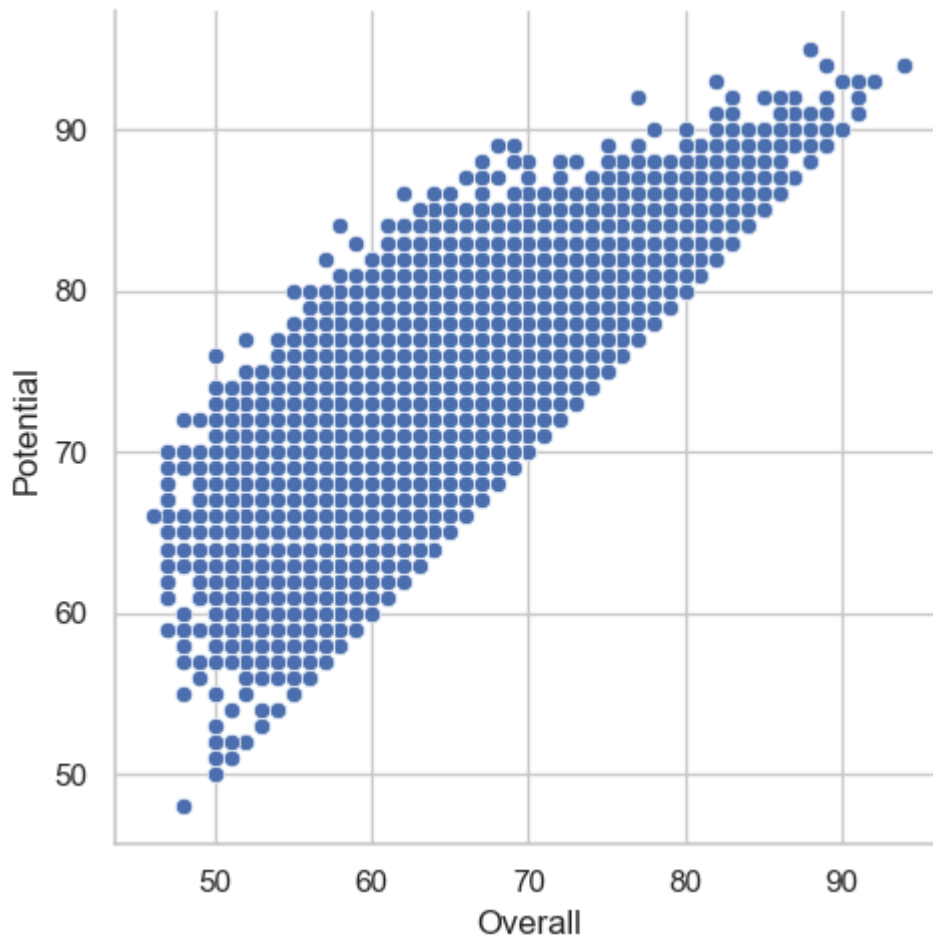
```
In [141... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show())
```



```
In [142... f,ax=plt.subplots(figsize=(10,7))
sns.barplot(x='International Reputation', y='Potential', data=fifa_data, palette
plt.show())
```

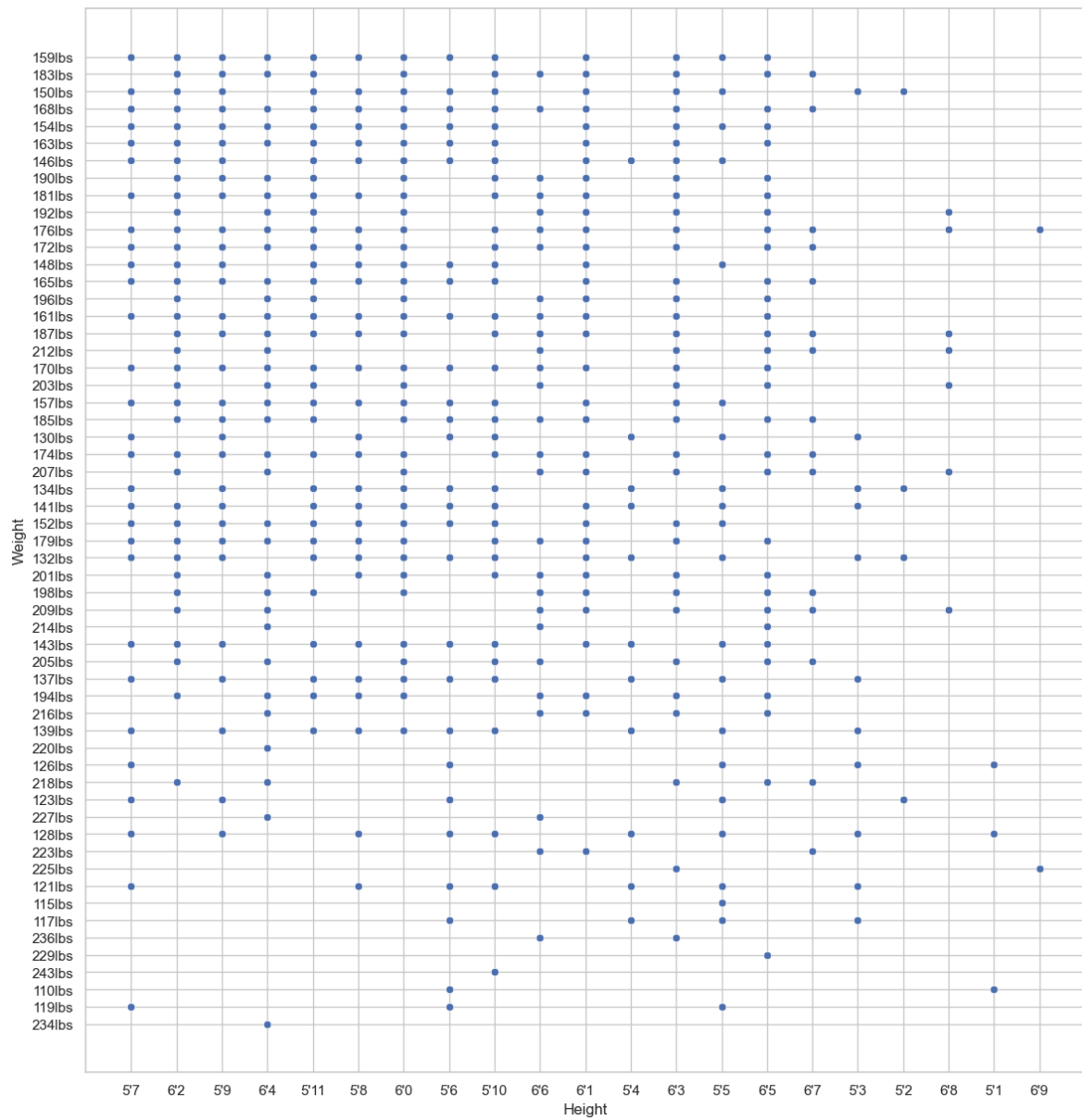


```
In [143... a = sns.relplot(x='Overall', y='Potential', data=fifa_data)#relplot is a scatter
```

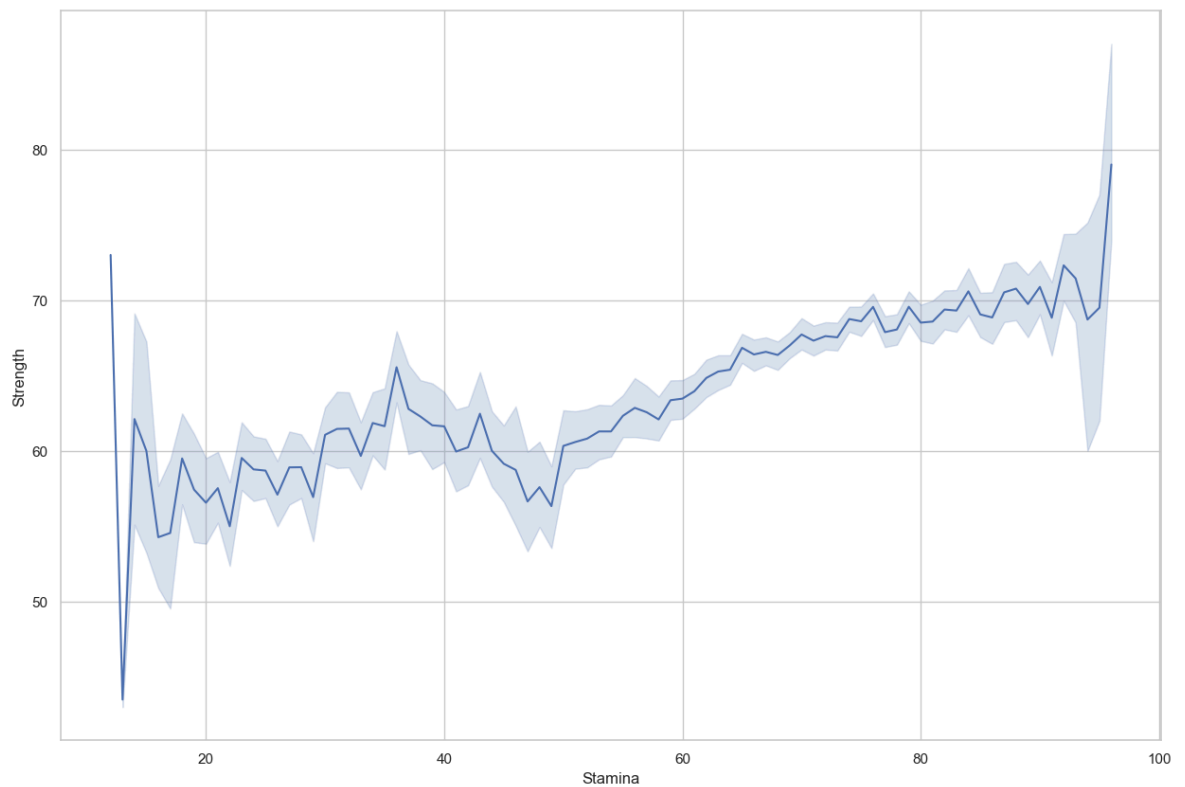


```
In [144... f,ax=plt.subplots(figsize=(14,15))
sns.scatterplot(x='Height', y='Weight', data=fifa_data, palette='Set2')
```

```
Out[144... <Axes: xlabel='Height', ylabel='Weight'>
```

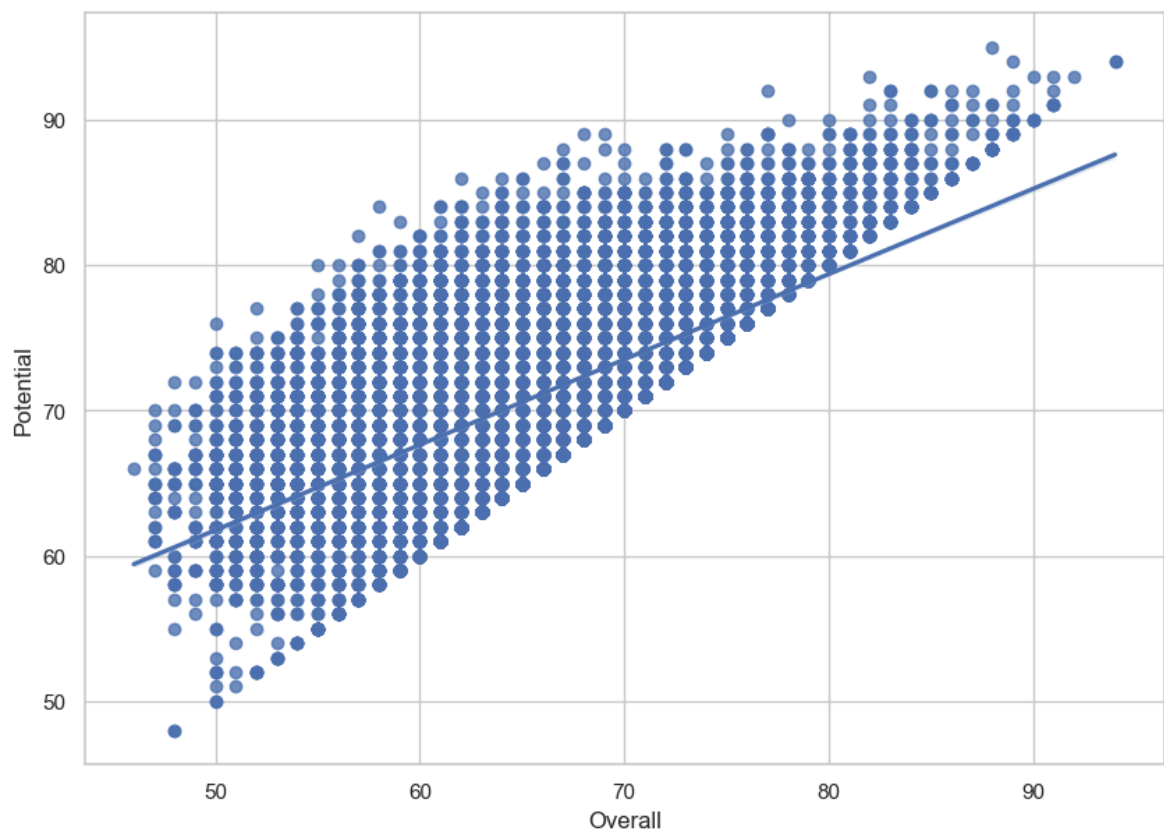


```
In [145... f,ax=plt.subplots(figsize=(15,10))
ax = sns.lineplot(x='Stamina', y='Strength', data=fifa_data)
plt.show()
```



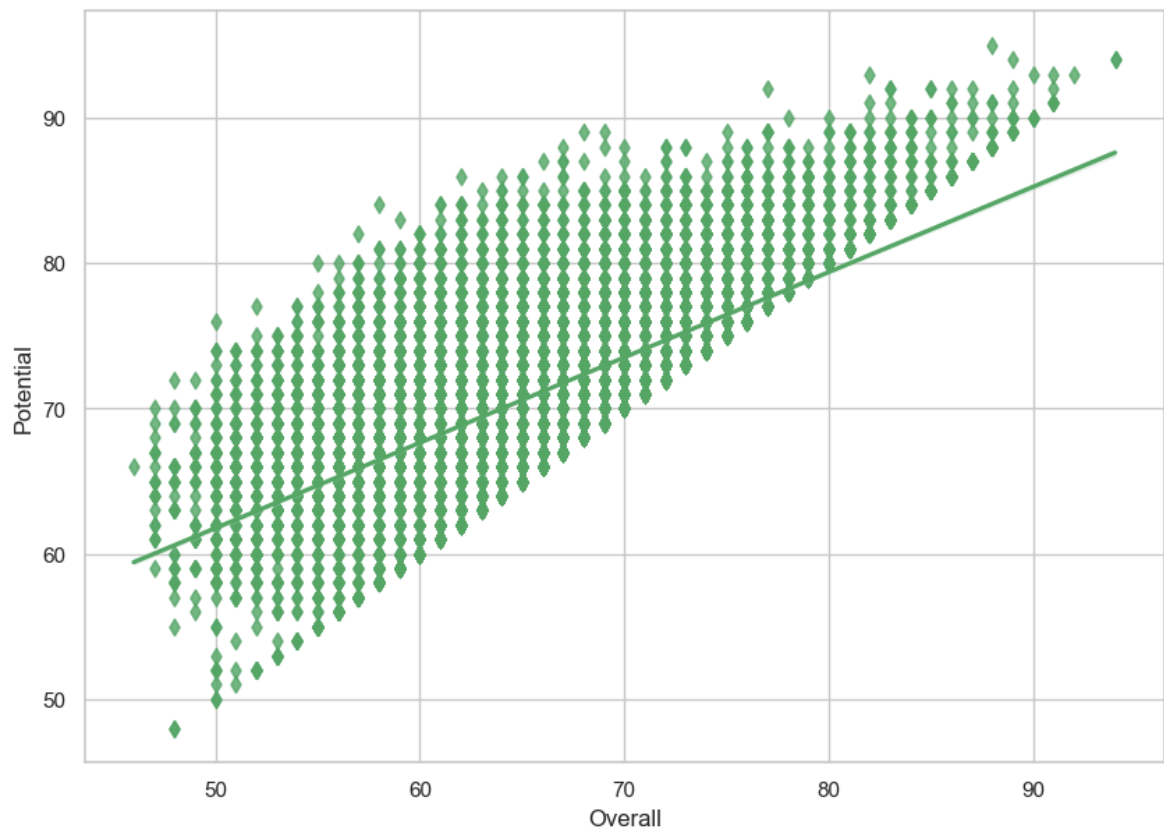
In [146...

```
f, ax=plt.subplots(figsize=(10,7))
ax = sns.regplot(x='Overall', y='Potential', data=fifa_data)#regplot is a scatter
plt.show()
```

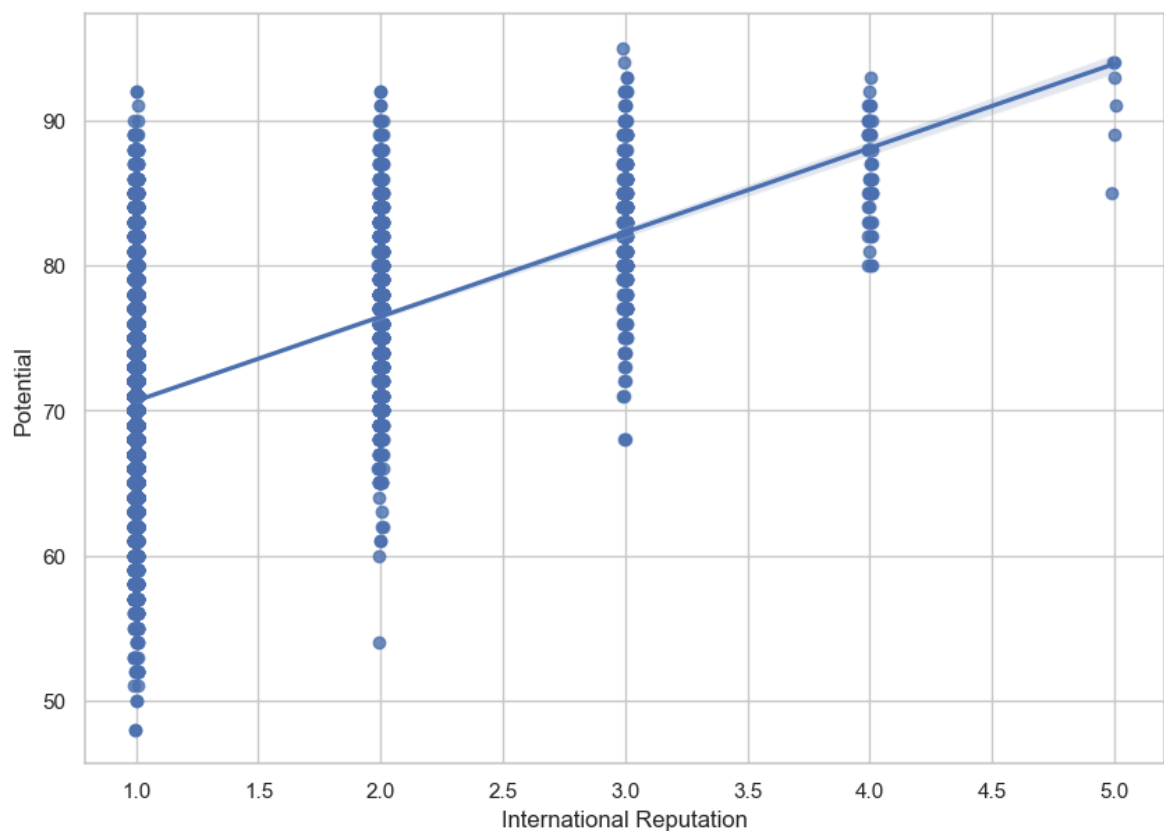


In [147...

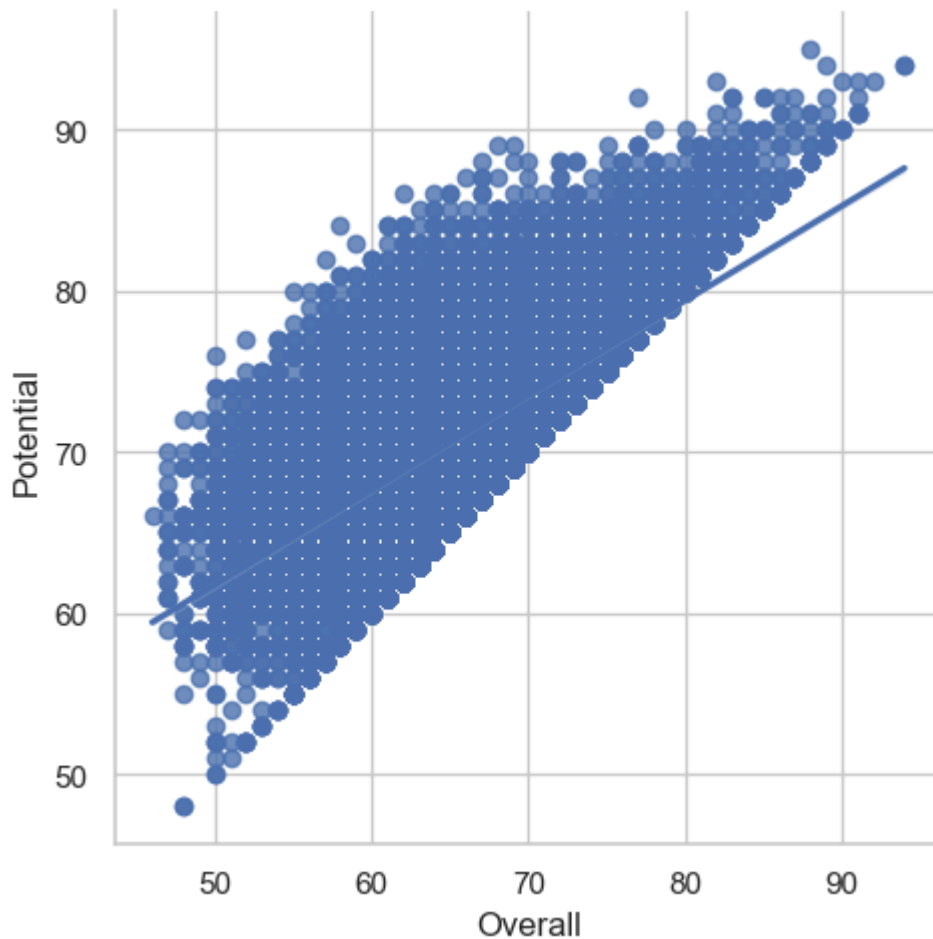
```
f, ax=plt.subplots(figsize=(10,7))
ax = sns.regplot(x='Overall', y='Potential', data=fifa_data, color='g', marker='o')
plt.show()
```



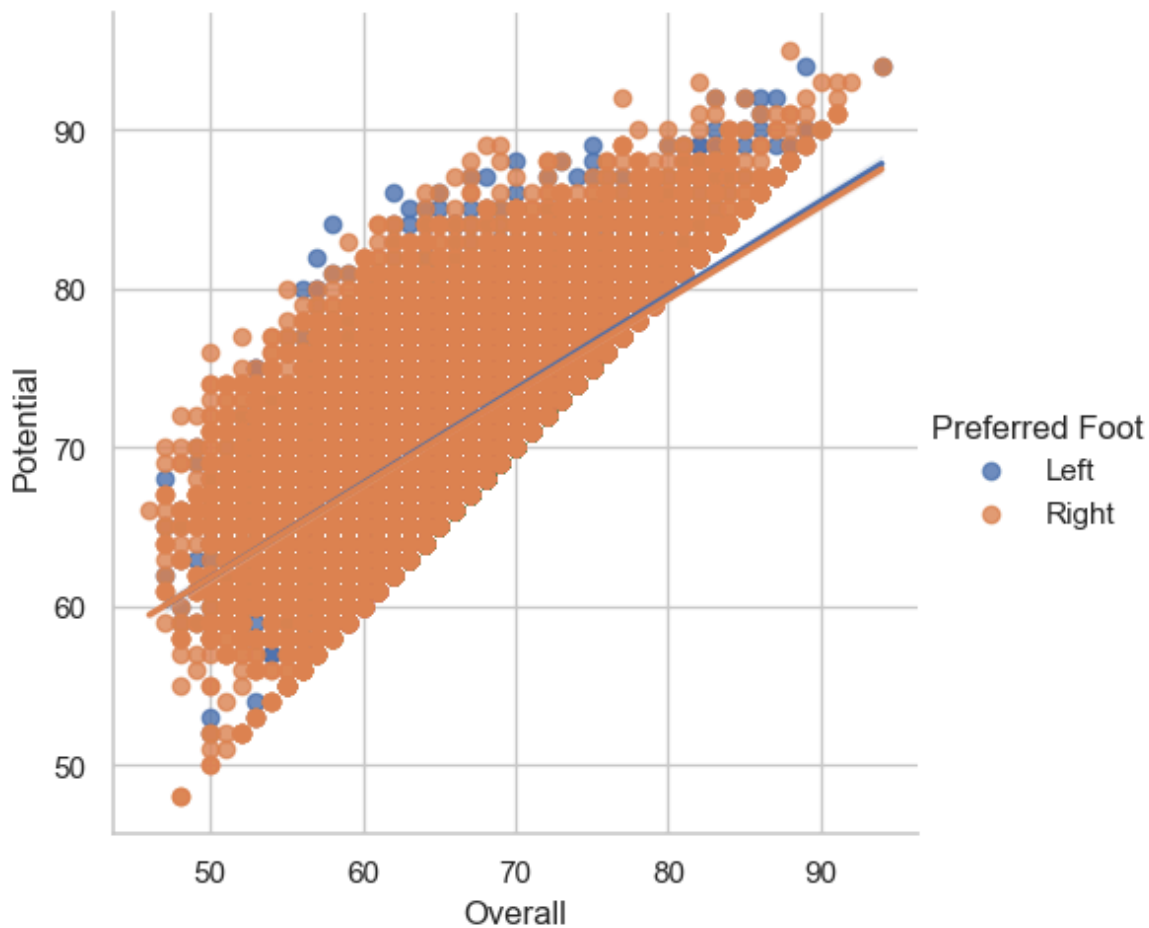
```
In [148... f, ax=plt.subplots(figsize=(10,7))
ax = sns.regplot(x='International Reputation', y='Potential', data=fifa_data, x_
plt.show())
```



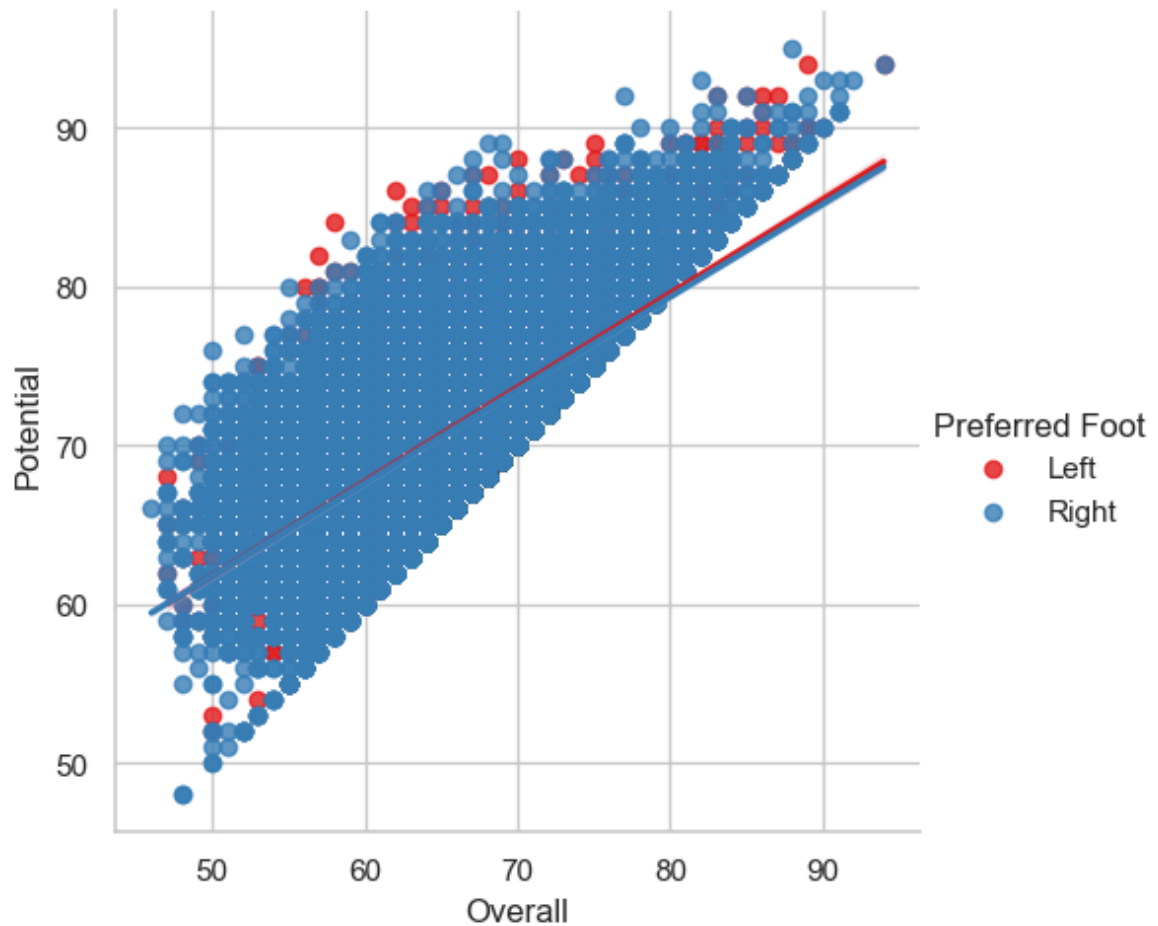
```
In [149... g = sns.lmplot(x="Overall", y="Potential", data=fifa_data) #lmplot is used to pl
```



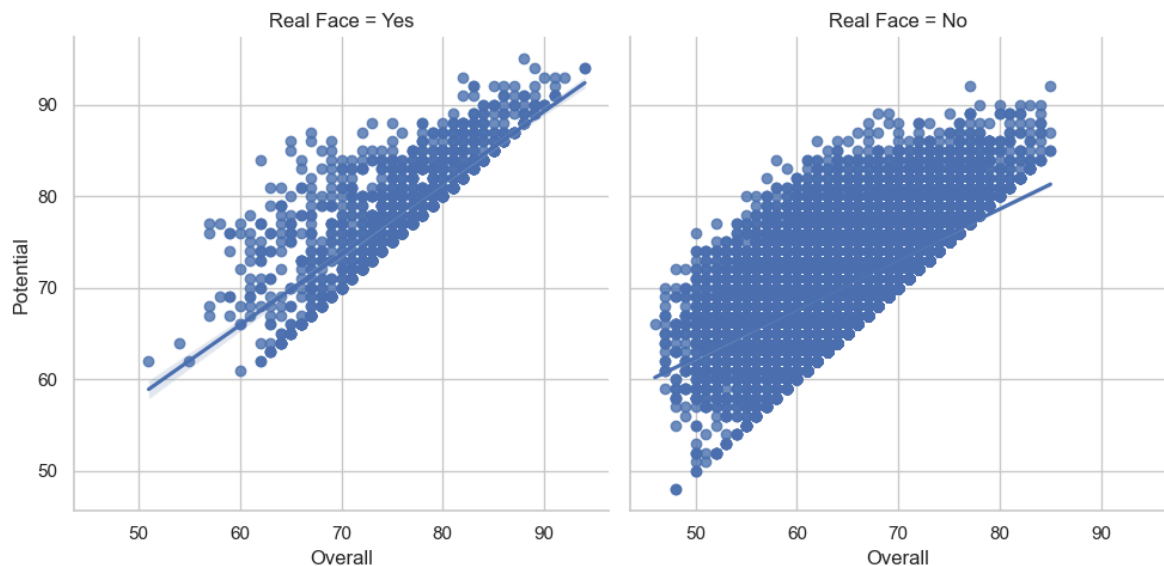
In [150... `g = sns.lmplot(x="Overall", y="Potential", data=fifa_data, hue='Preferred Foot')`



```
In [151... g = sns.lmplot(x="Overall", y="Potential", data=fifa_data, hue='Preferred Foot',
```

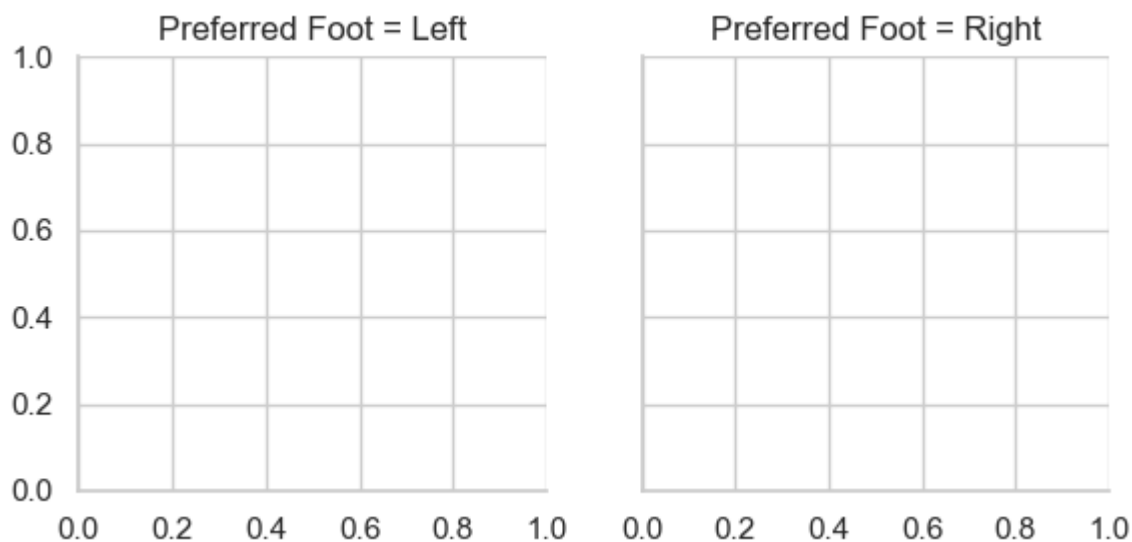


```
In [152... g = sns.lmplot(x="Overall", y="Potential", data=fifa_data, col='Real Face')
```

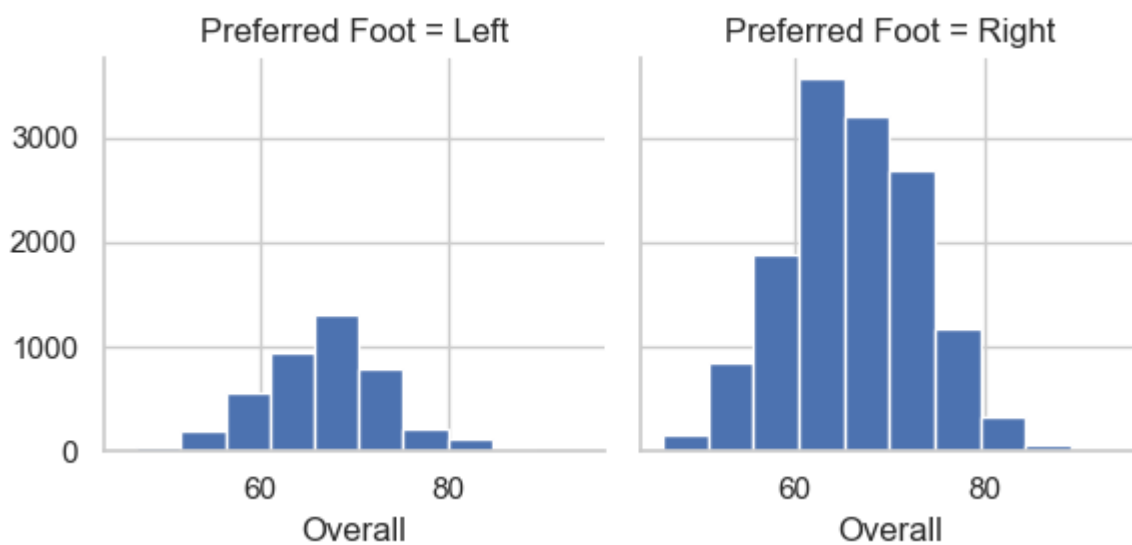


Multi-plot Grids

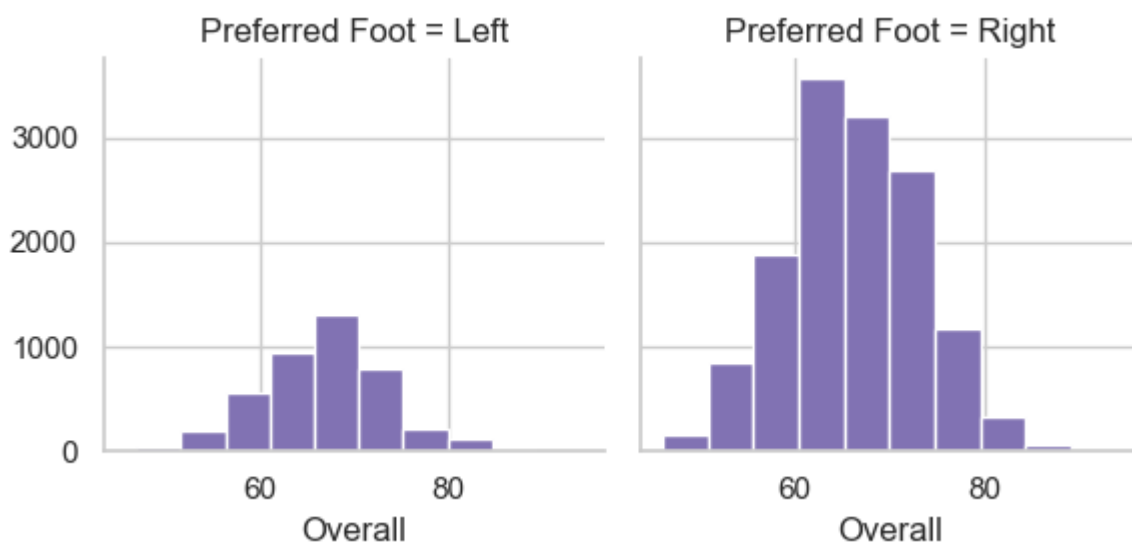
```
In [153... g = sns.FacetGrid(fifa_data, col="Preferred Foot")
```



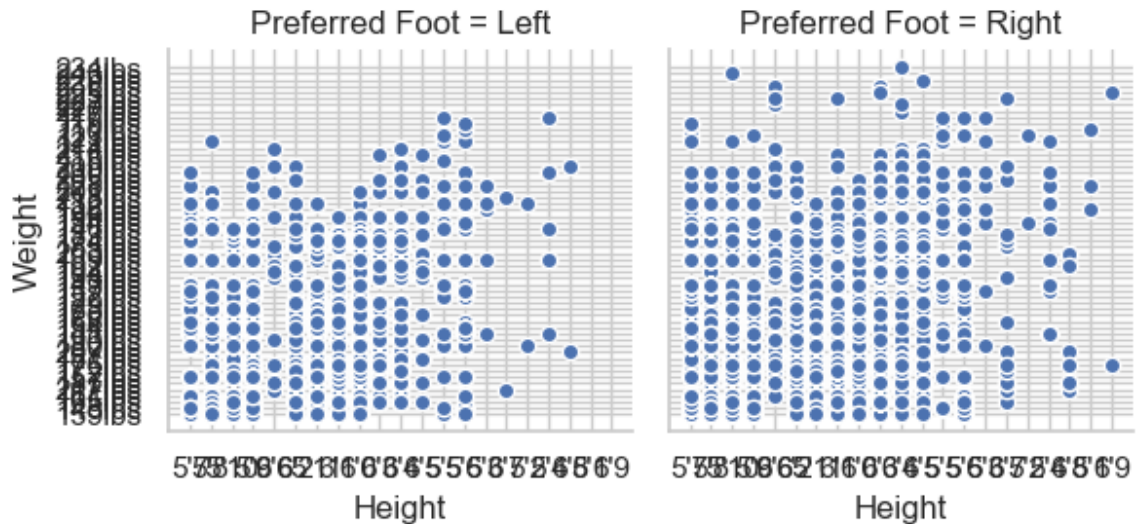
```
In [154... g = sns.FacetGrid(fifa_data, col="Preferred Foot")
g=g.map(plt.hist, "Overall")
```



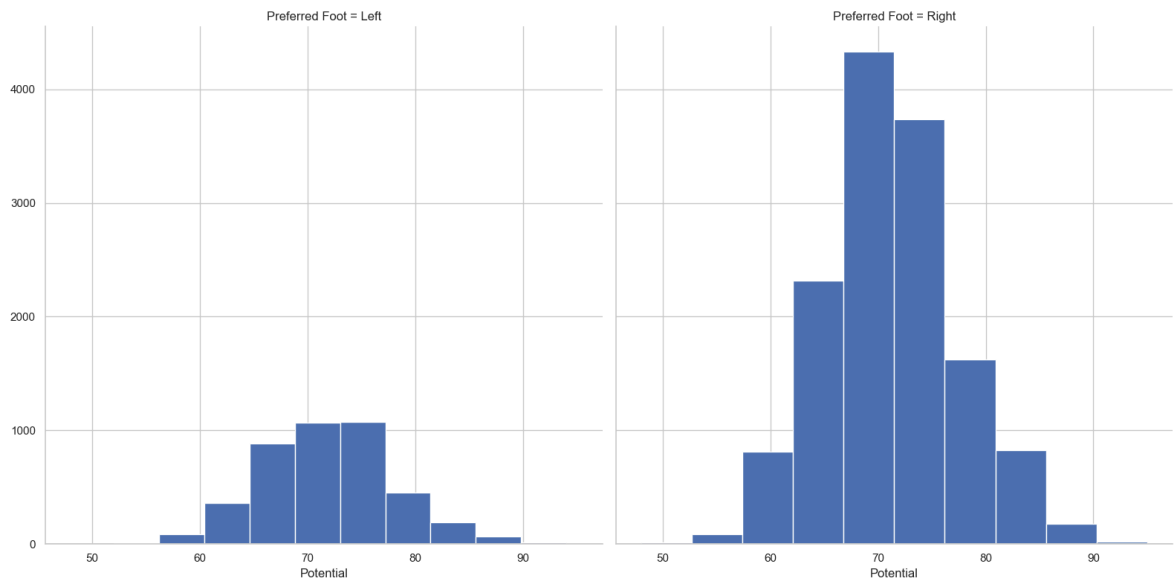
```
In [155... g = sns.FacetGrid(fifa_data, col="Preferred Foot")
g=g.map(plt.hist, "Overall", bins=10, color='m')
```



```
In [156... ax = sns.FacetGrid(fifa_data, col='Preferred Foot')
ax=(ax.map(plt.scatter, 'Height', 'Weight', edgecolor='w')).add_legend()
```



```
In [157... g = sns.FacetGrid(fifa_data, col="Preferred Foot", height=8, aspect=1)
g = g.map(plt.hist, "Potential")
```



```
In [158... fifa_data_new=fifa_data[['Age', 'Potential', 'Strength', 'Stamina', 'Preferred F
```

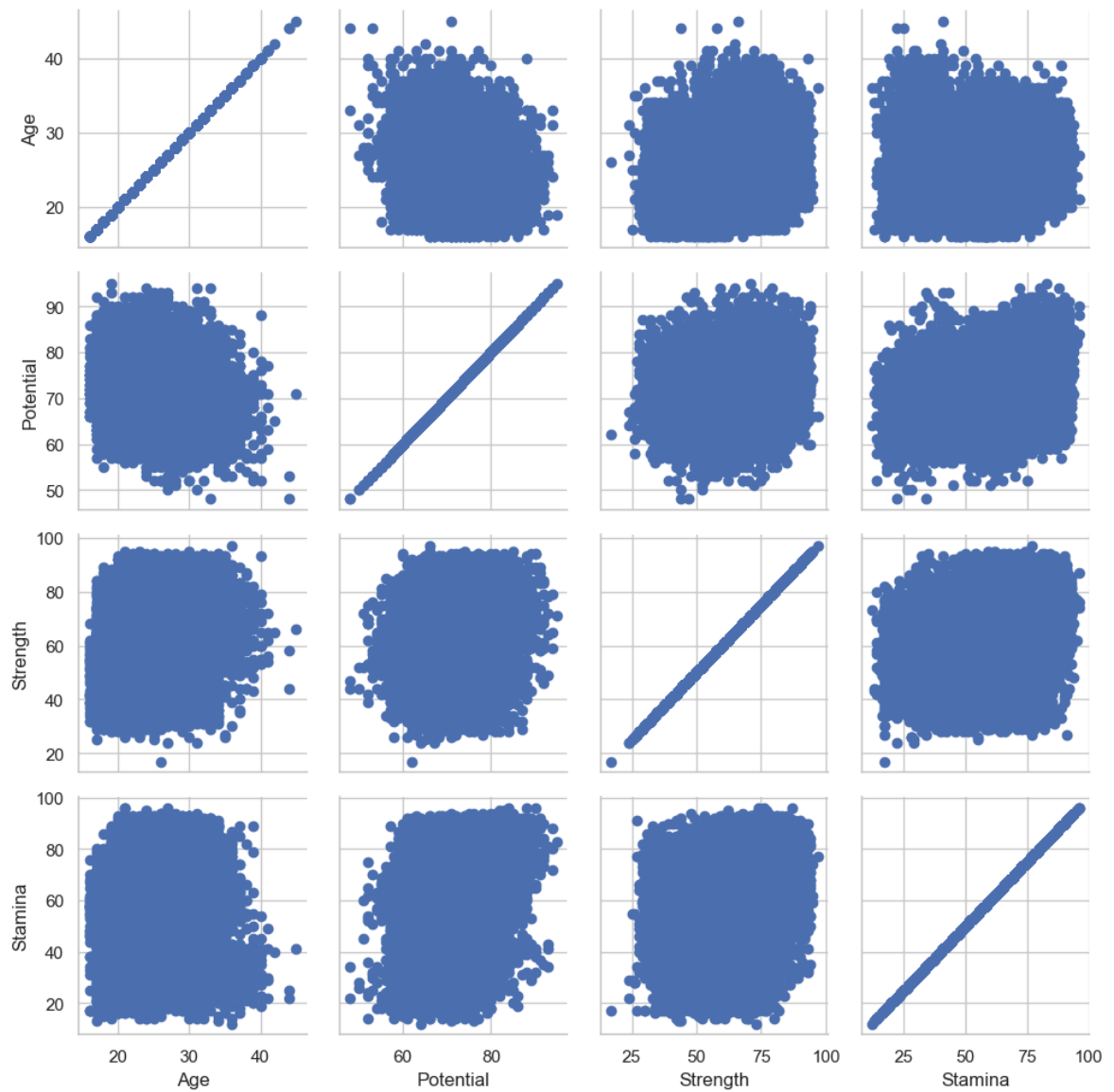
```
In [159... fifa_data_new.head()
```

```
Out[159... 
```

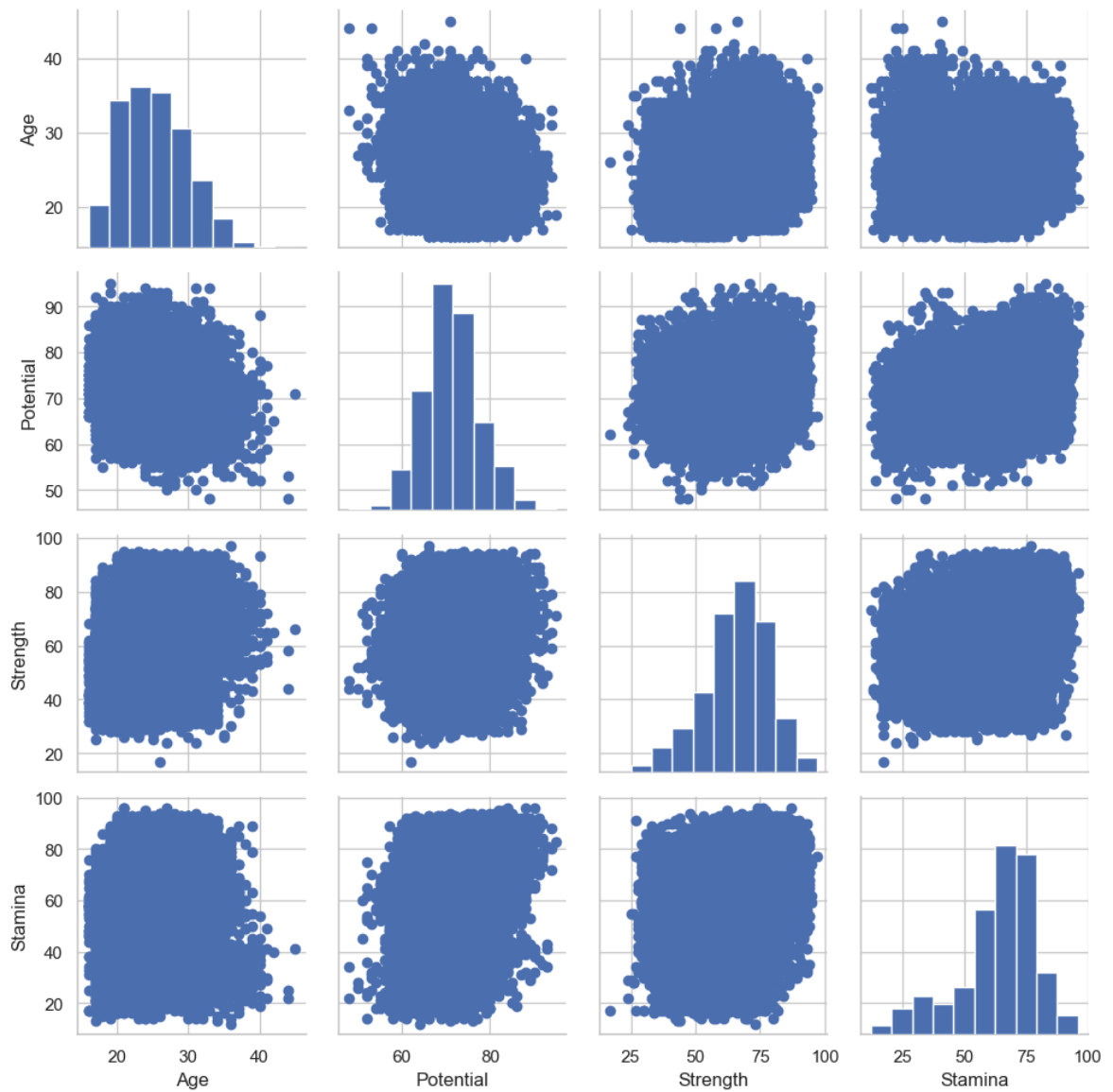
	Age	Potential	Strength	Stamina	Preferred Foot
0	31	94	59.0	72.0	Left
1	33	94	79.0	88.0	Right
2	26	93	49.0	81.0	Right
3	27	93	64.0	43.0	Right
4	27	92	75.0	90.0	Right

```
In [160... g=sns.PairGrid(fifa_data_new) #pairgrid shows how variables are related to each
#pair grid is used when we want to compare multiple variables at once.
```

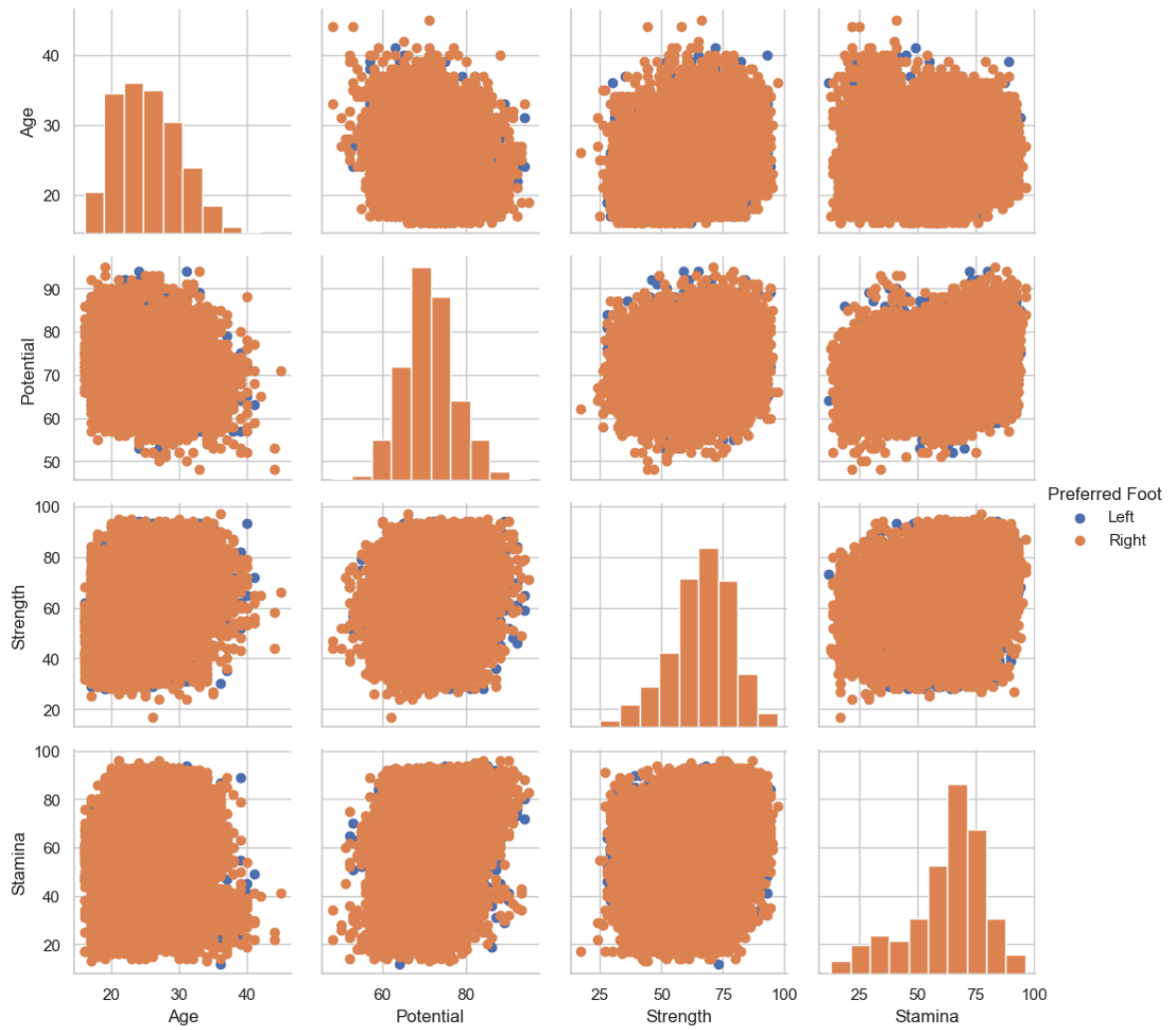
```
g=g.map(plt.scatter)
```



```
In [161... g = sns.PairGrid(fifa_data_new)
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
```

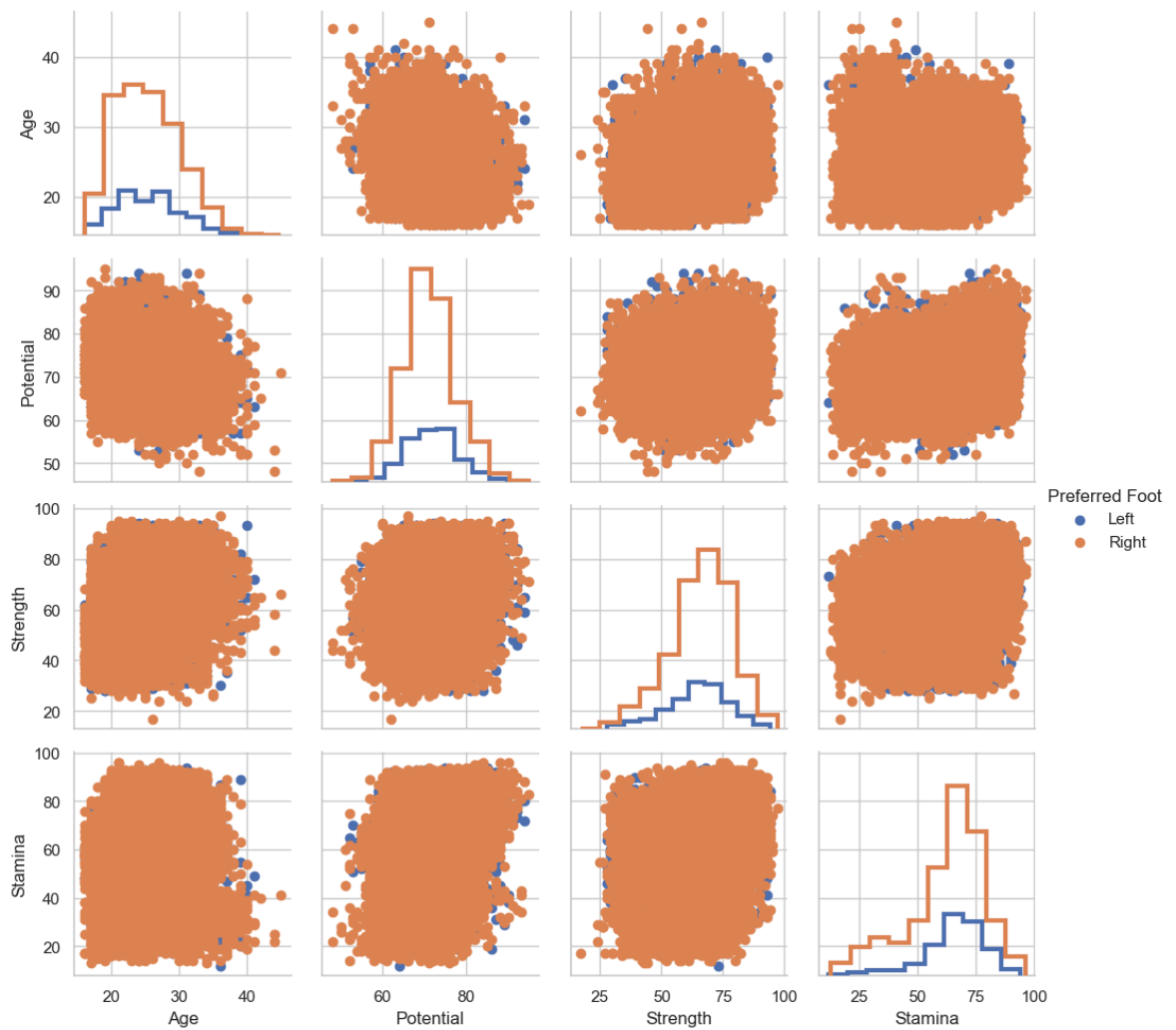


```
In [162... g = sns.PairGrid(fifa_data_new, hue='Preferred Foot')
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
```



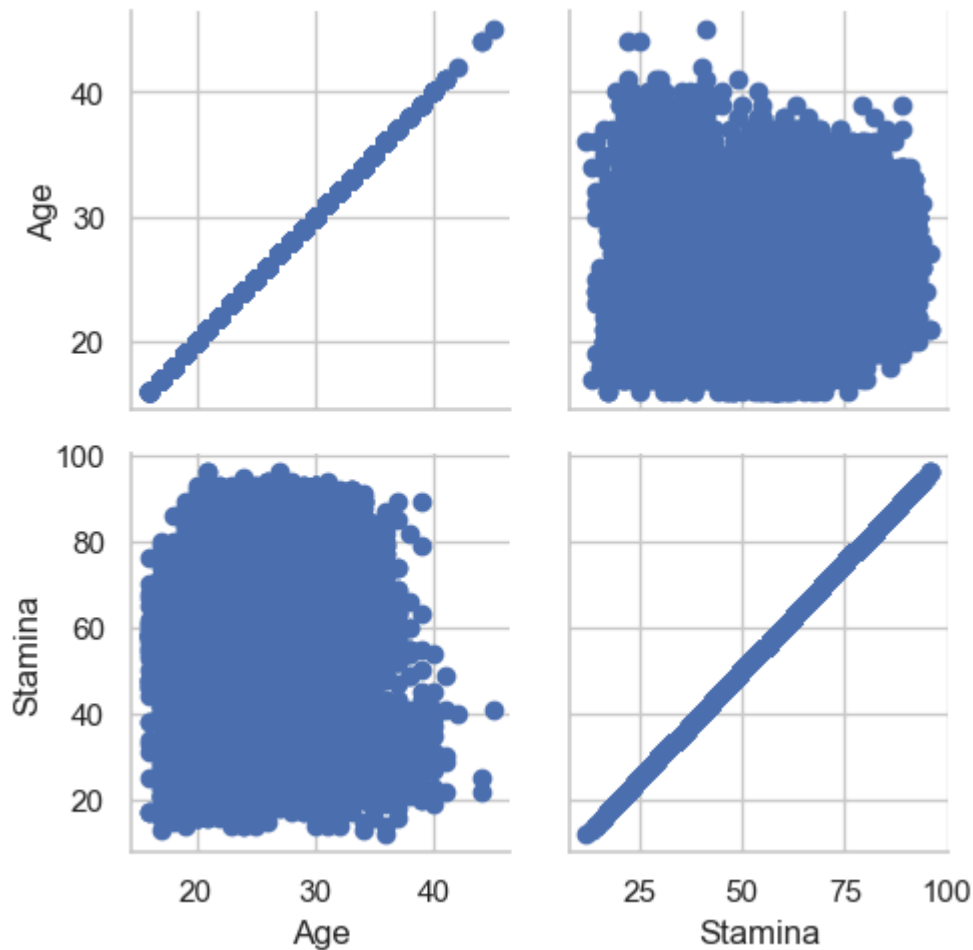
In [163...

```
g = sns.PairGrid(fifa_data_new, hue='Preferred Foot')
g = g.map_diag(plt.hist, histtype='step', linewidth=3)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
```



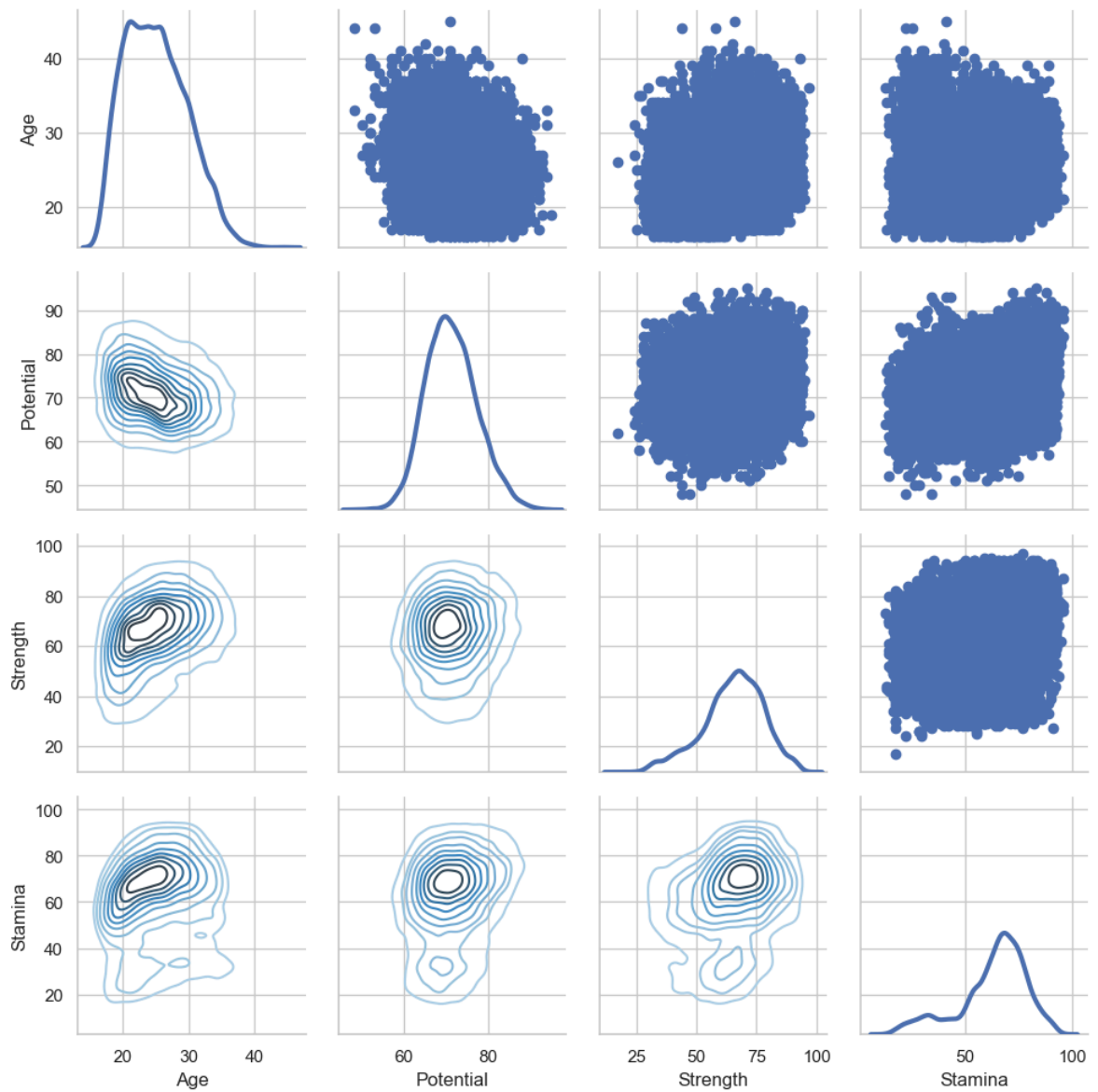
In [164...

```
g = sns.PairGrid(fifa_data_new, vars=['Age', 'Stamina'])  
g = g.map(plt.scatter)
```

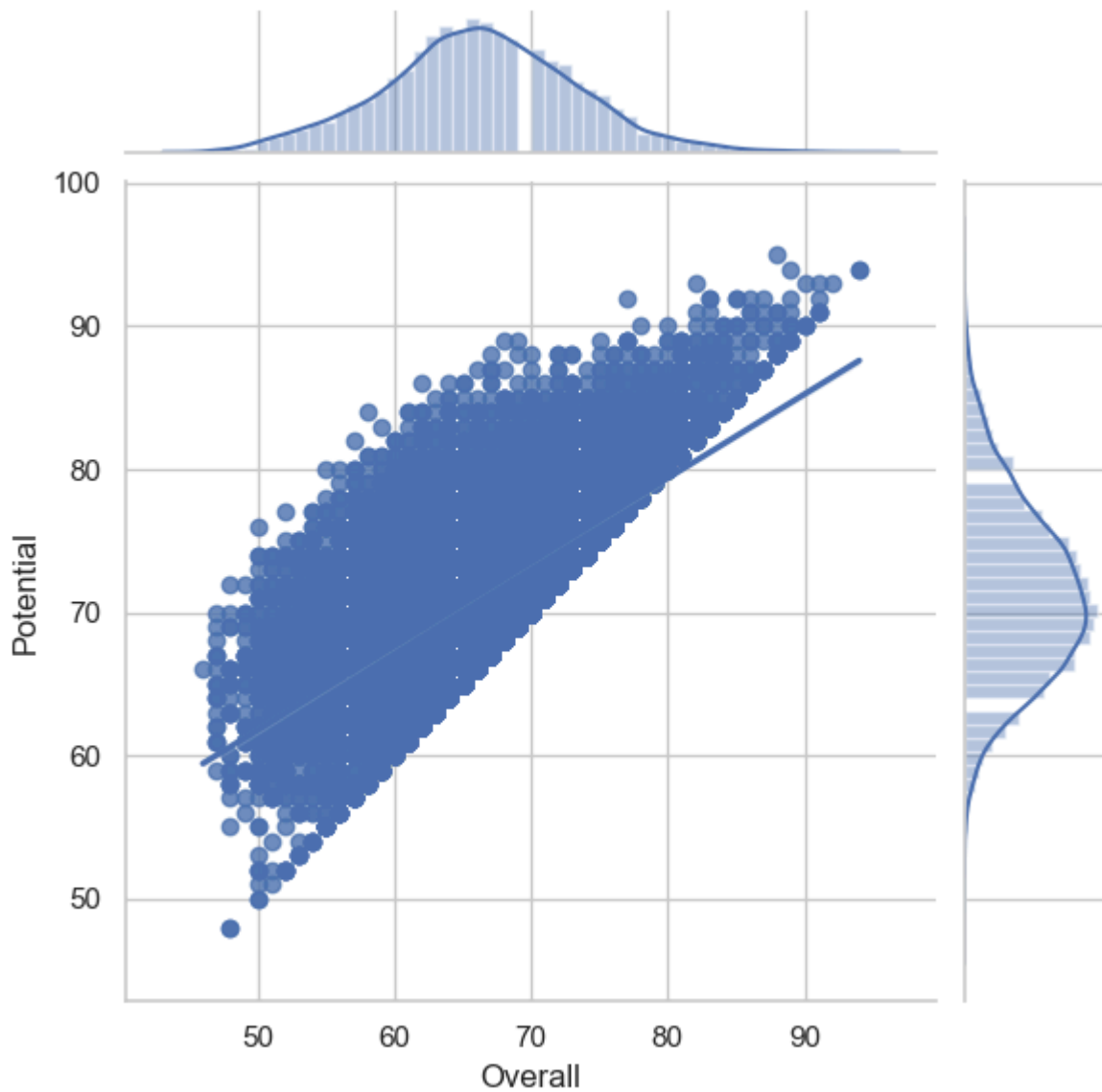


In [165...

```
g = sns.PairGrid(fifa_data_new)
g = g.map_upper(plt.scatter) #in upper triangle we plot scatter plot
g = g.map_lower(sns.kdeplot, cmap="Blues_d") # in lower triangle we plot kde plo
g = g.map_diag(sns.kdeplot, lw=3, legend=False) #we are getting line cuz of lw=3
```

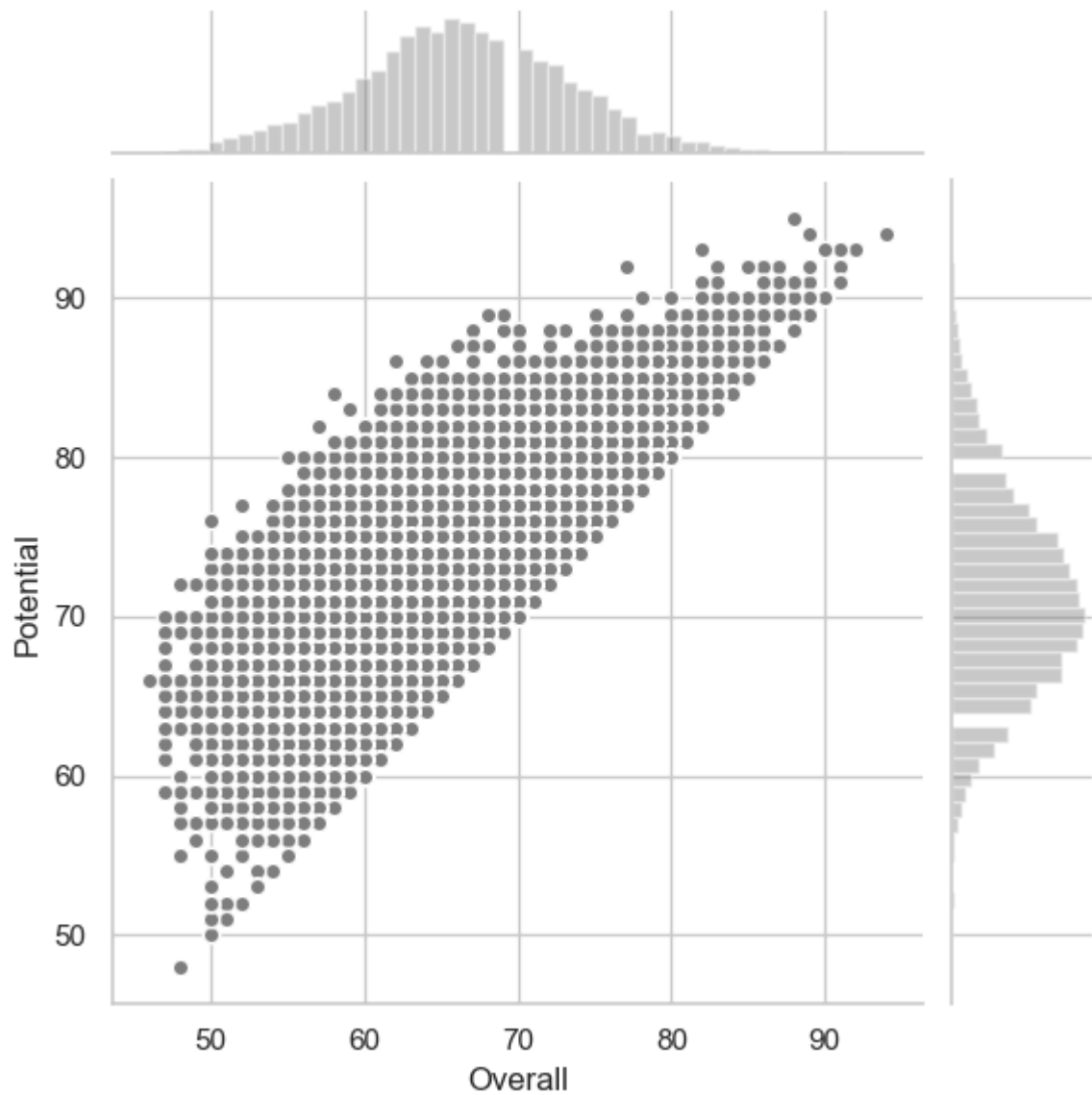


```
In [166... g = sns.JointGrid(x="Overall", y="Potential", data=fifa_data)
g = g.plot(sns.regplot, sns.distplot)
```

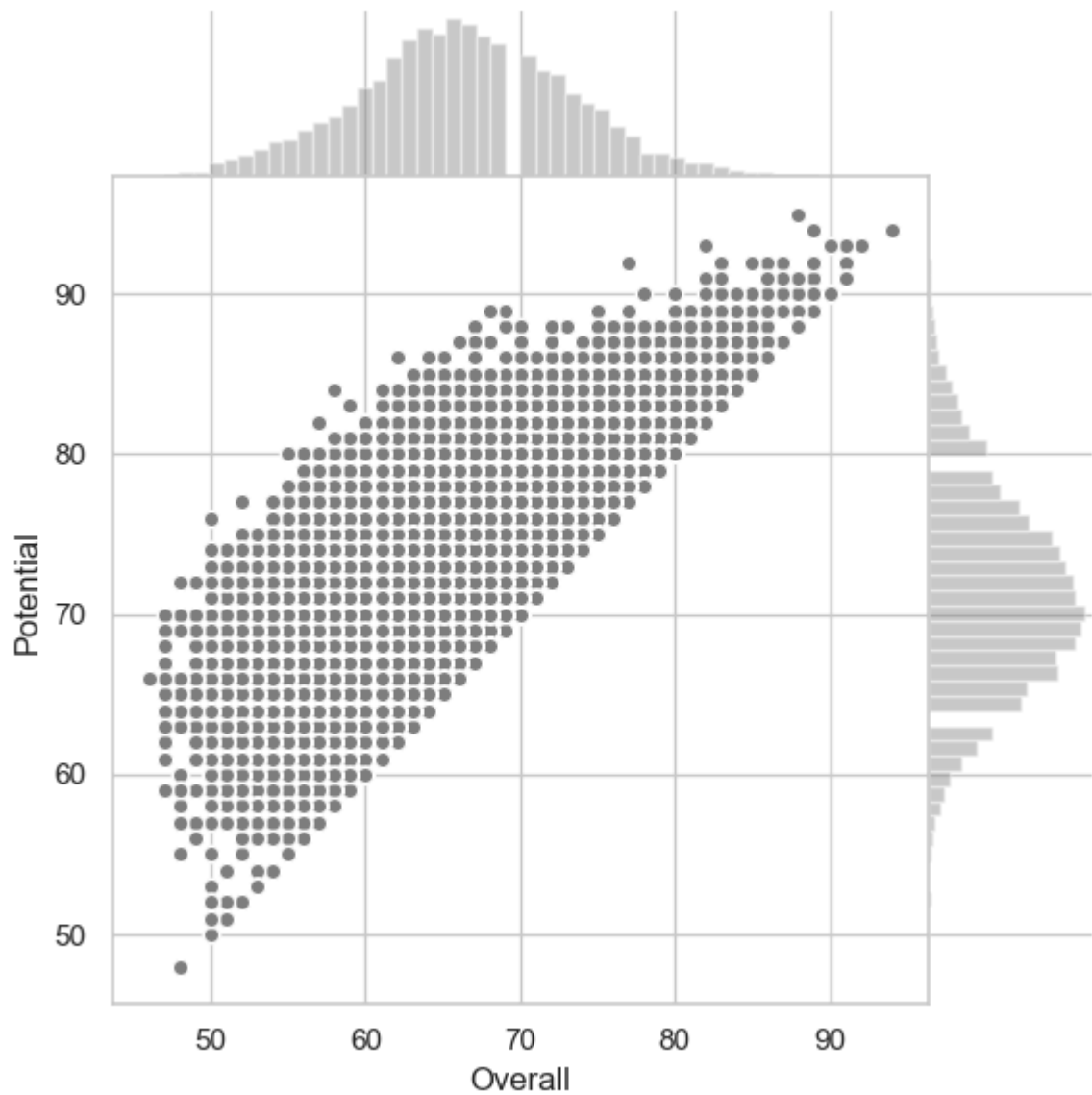


In [167...

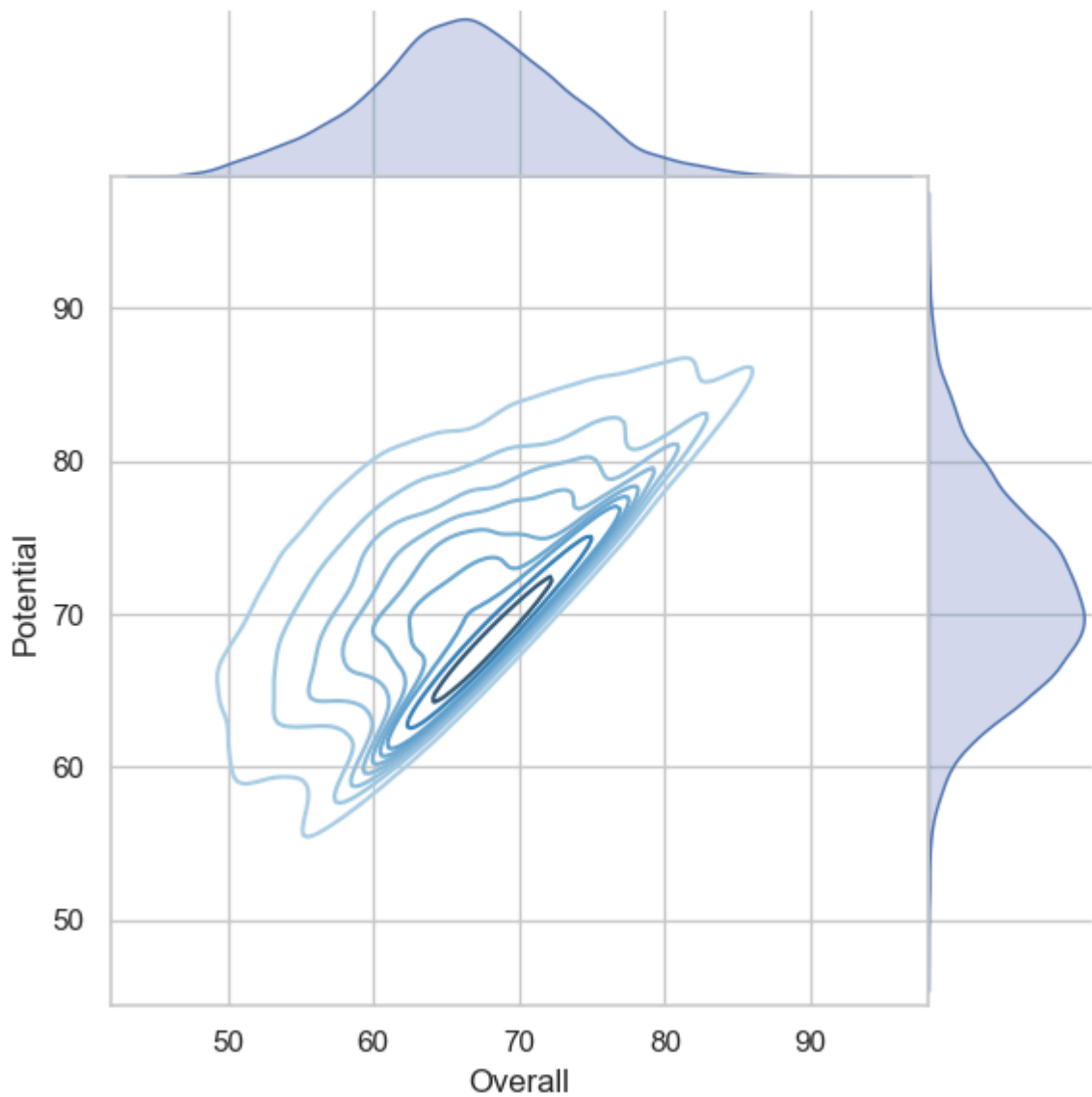
```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa_data)
g = g.plot_joint(plt.scatter, color=".5", edgecolor="white")
g = g.plot_marginals(sns.distplot, kde=False, color=".5")
```



```
In [168... g = sns.JointGrid(x="Overall", y="Potential", data=fifa_data, space=0)#space is
g = g.plot_joint(plt.scatter, color=".5", edgecolor="white")
g = g.plot_marginals(sns.distplot, kde=False, color=".5")
```

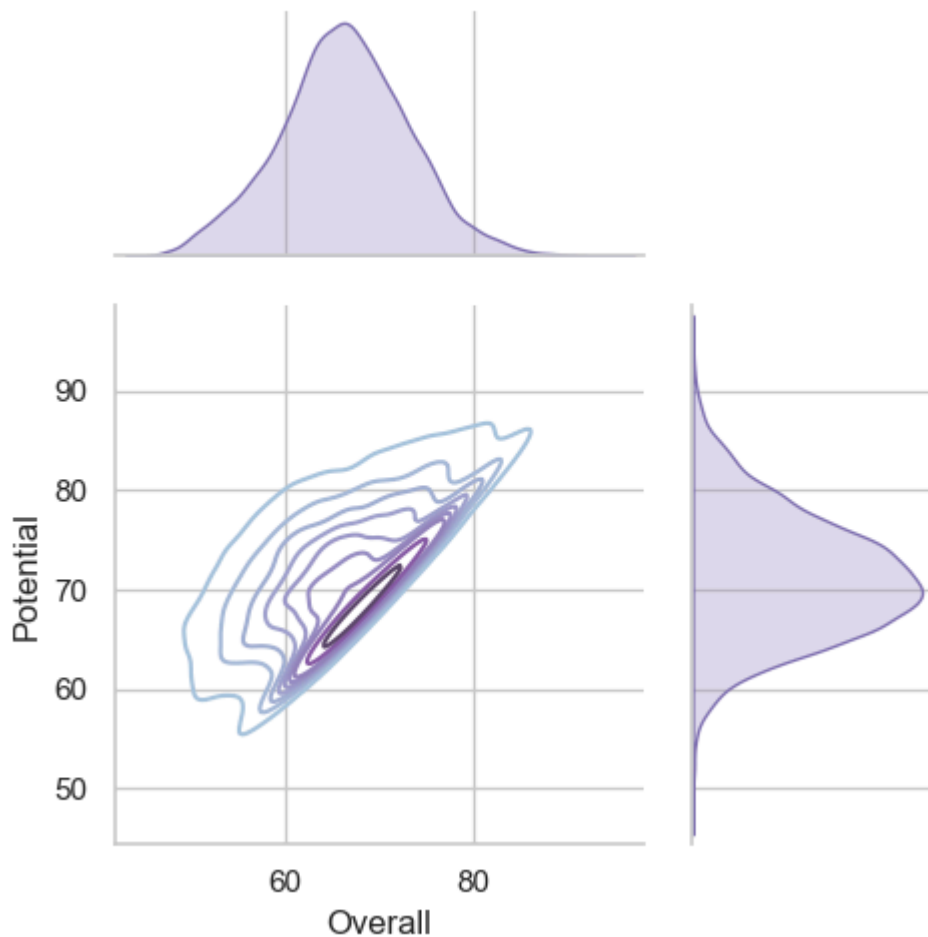


```
In [169... g = sns.JointGrid(x="Overall", y="Potential", data=fifa_data, space=0)
g = g.plot_joint(sns.kdeplot, cmap="Blues_d")
g = g.plot_marginals(sns.kdeplot, shade=True)
```

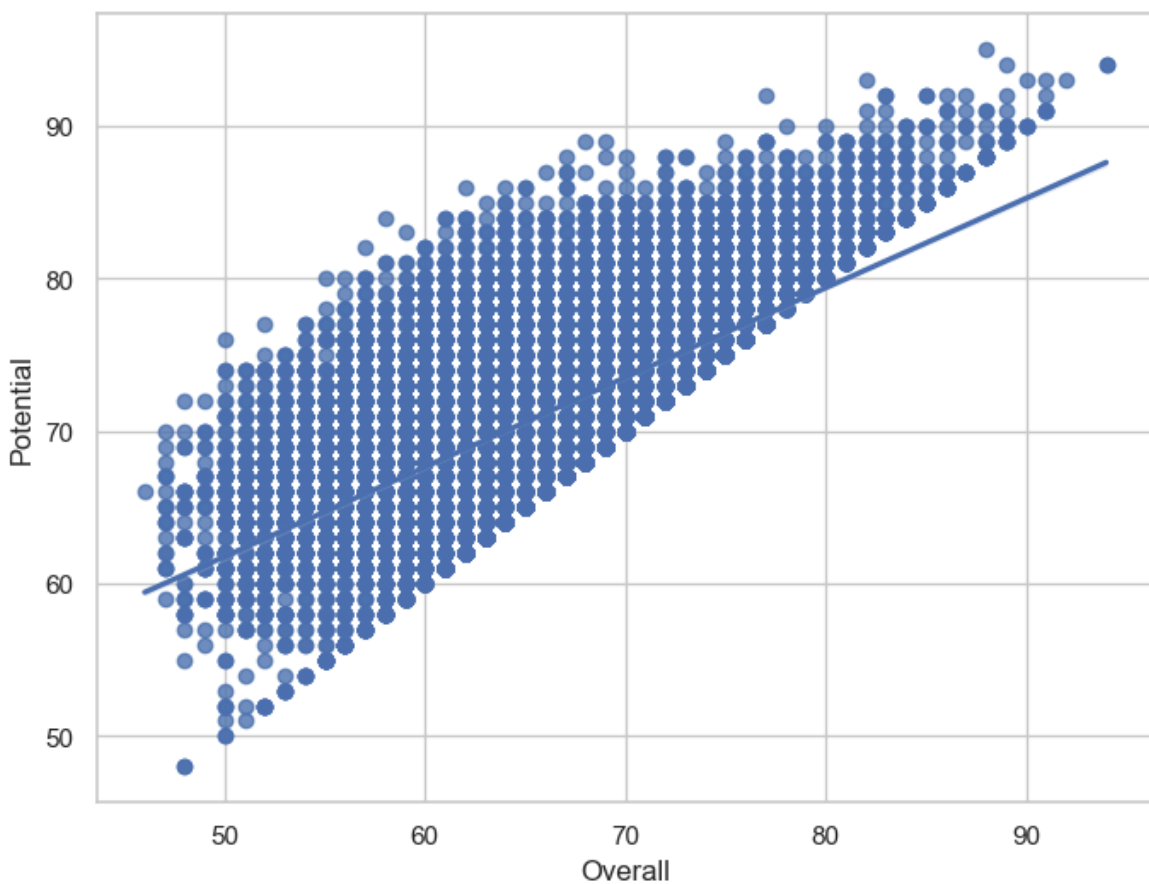


In [172...

```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa_data, height=5, ratio=2)
g = g.plot_joint(sns.kdeplot, cmap="BuPu_d")
g = g.plot_marginals(sns.kdeplot, shade=True, color='m')
```

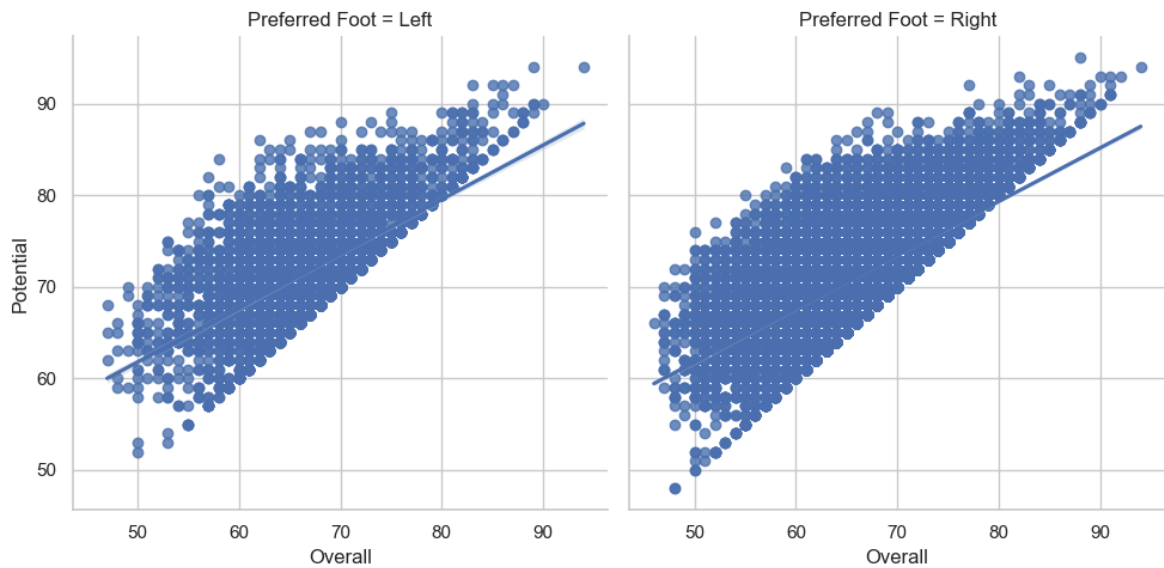


```
In [173... f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="Overall", y="Potential", data=fifa_data)
```



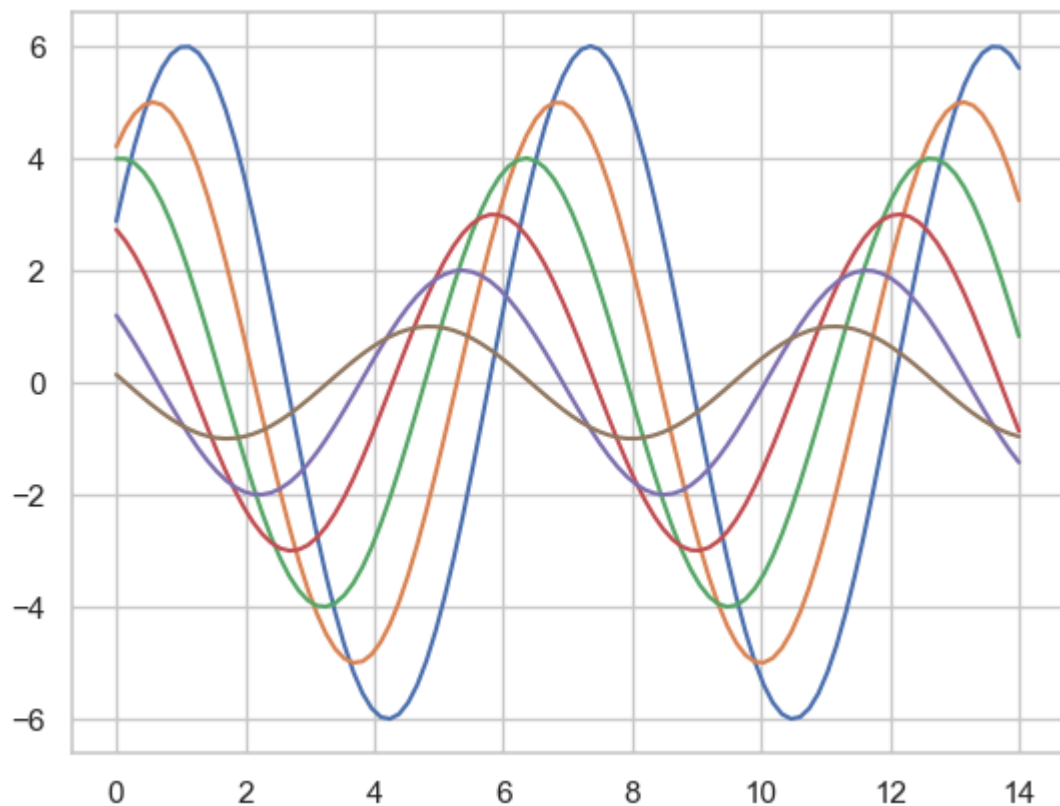
In [174... `sns.lmplot(x="Overall", y="Potential", col="Preferred Foot", data=fifa_data, col`

Out[174... `<seaborn.axisgrid.FacetGrid at 0x25ce3f87050>`

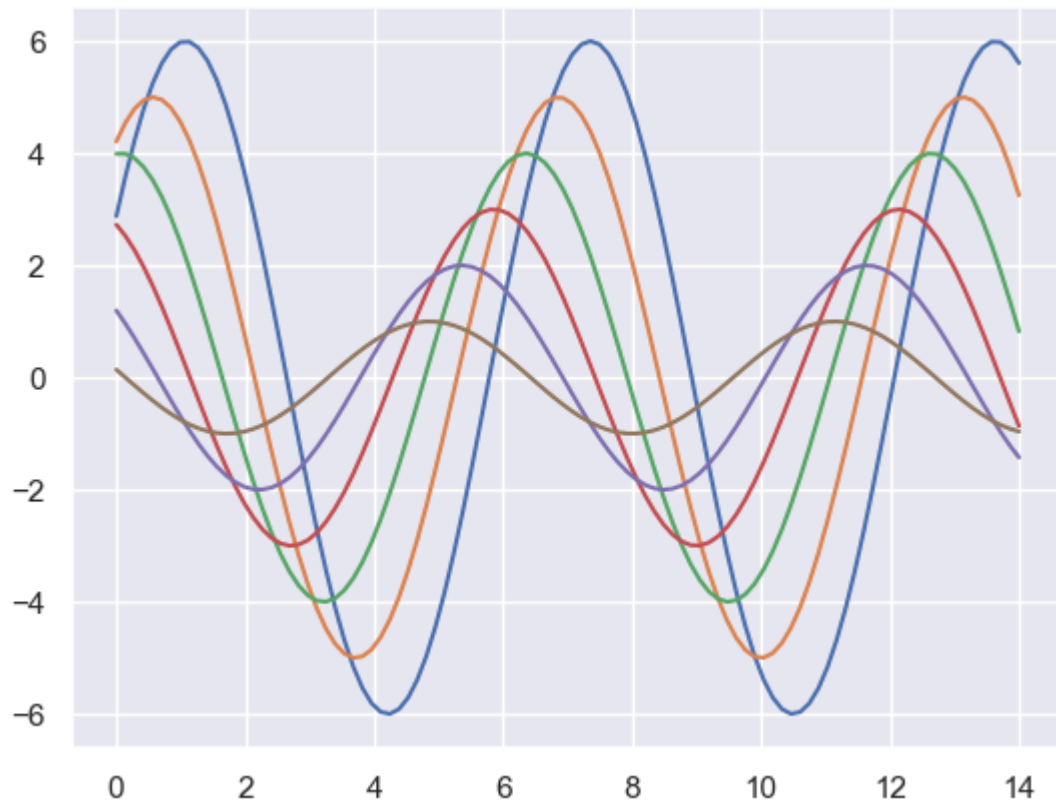


```
In [175... def sinplot(flip=1):
    x = np.linspace(0, 14, 100)
    for i in range(1, 7):
        plt.plot(x, np.sin(x + i * .5) * (7 - i) * flip)
```

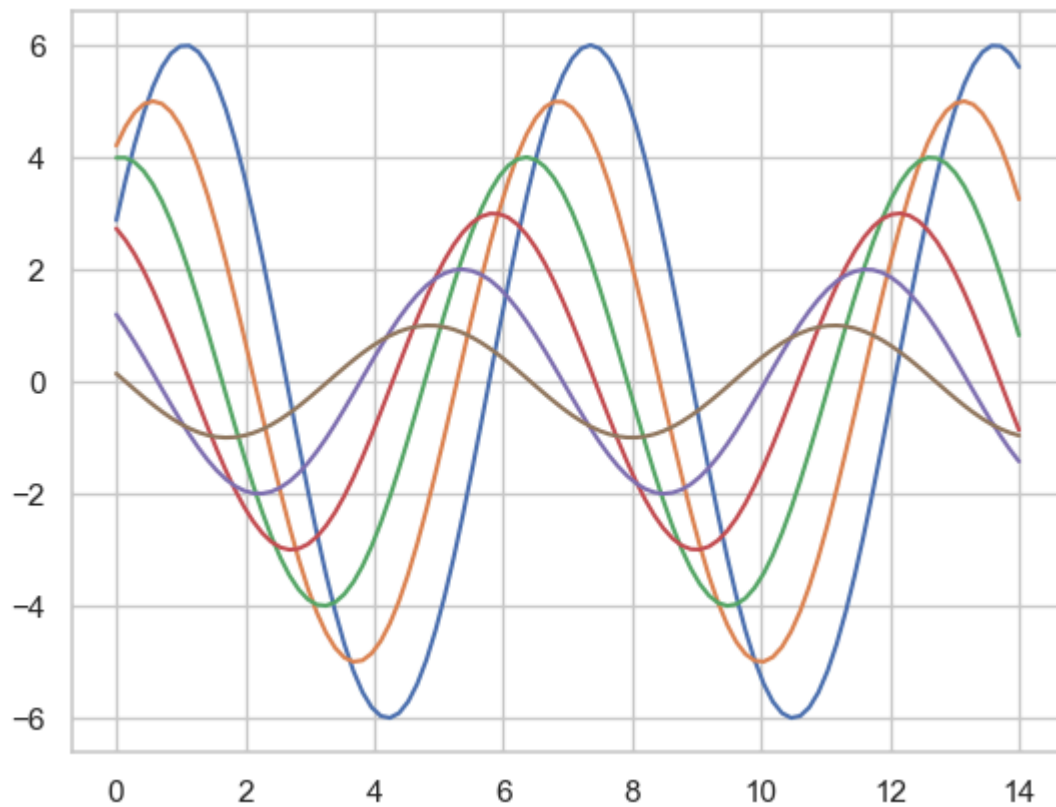
In [176... `sinplot()`



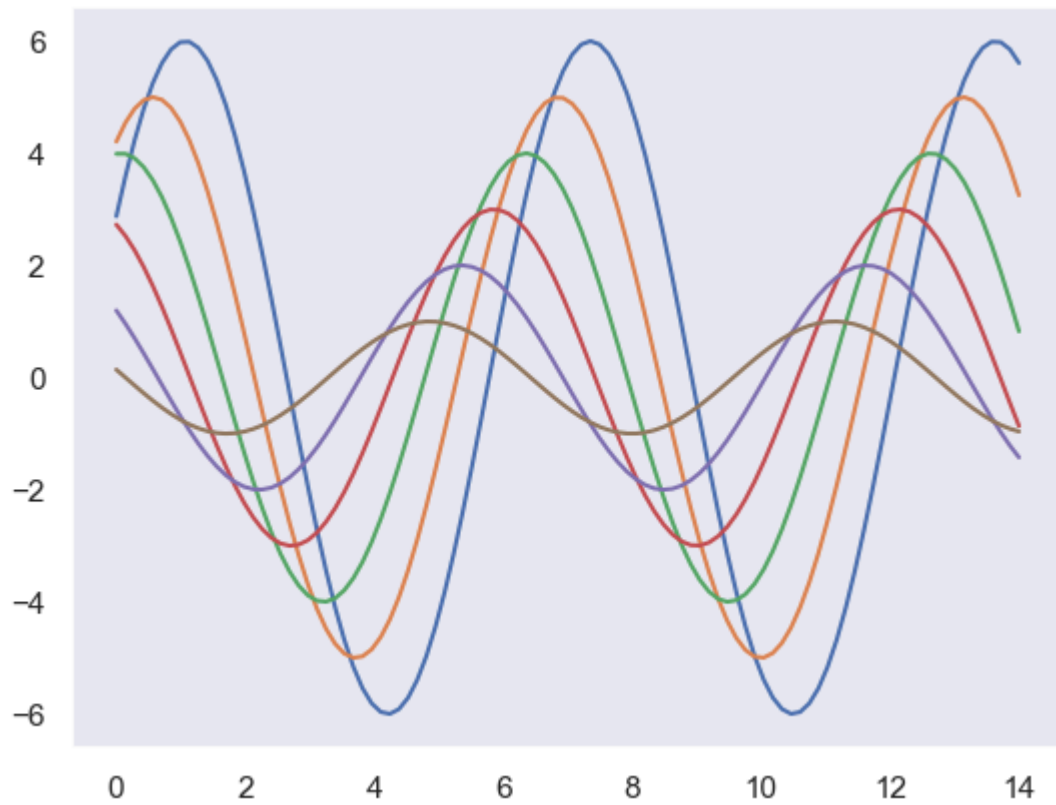
```
In [ ]: sns.set()#set() is used to switch to seaborn's default style
sinplot()
```



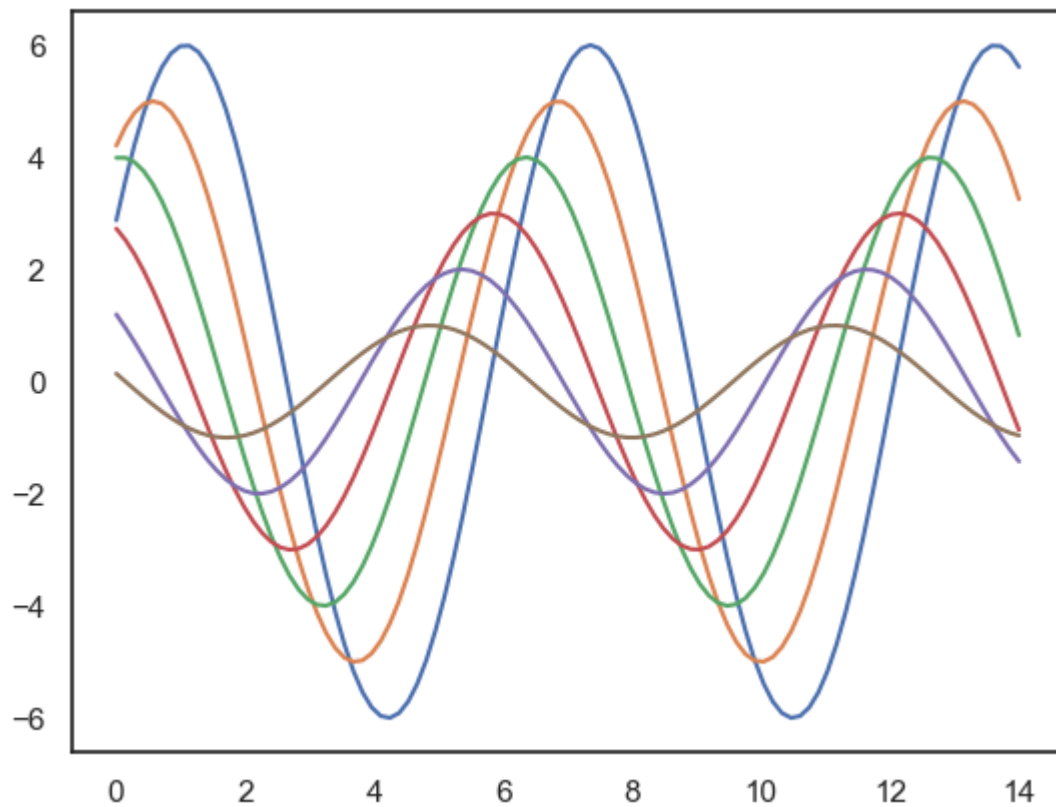
```
In [178... sns.set_style("whitegrid")  
sinplot()
```



```
In [179... sns.set_style("dark")  
sinplot()
```



```
In [180... sns.set_style("white")  
sinplot()
```



```
In [ ]: sns.set_style("ticks")  
sinplot()
```

